

KAZAKH-BRITISH TECHNICAL UNIVERSITY

School of Information Technology

**DEVELOPMENT OF AN INTELLIGENT SYSTEM FOR
INVESTIGATING AND SOLVING ECOLOGICAL PROBLEMS**

**РАЗРАБОТКА ИНТЕЛЛЕКТУАЛЬНОЙ СИСТЕМЫ ДЛЯ
ИССЛЕДОВАНИЯ И РЕШЕНИЯ ЭКОЛОГИЧЕСКИХ ПРОБЛЕМ**

Master's Thesis

Programme 7M06106 "Software Engineering"

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0.1. ABSTRACT / АННОТАЦИЯ / АНДАТПА

0.1.1. ABSTRACT (English)

Development of an intelligent system for investigating and solving ecological problems

Master's thesis - Program 7M06106 "Software Engineering", KBTU, 2026
Supervisor: [Supervisor full name, academic title]

The relevance of the present work is determined by the acute public health challenge of wintertime PM_{2.5} air pollution in Almaty, Kazakhstan. The city's mountain-basin topography and coal-dependent private-sector heating combine to produce seasonal inversion episodes in which the atmospheric boundary layer collapses to 10–50 m and daily-mean PM_{2.5} concentrations reach 74–81 $\mu\text{g}/\text{m}^3$ - five to six times the WHO 24-hour guideline of 15 $\mu\text{g}/\text{m}^3$. Despite this severity, no open, reproducible, locally calibrated short-term PM_{2.5} forecast has been available to residents or municipal agencies.

The **aim** of the work is to develop and evaluate an intelligent short-term PM_{2.5} forecasting system for Almaty built entirely on open data and deployable on free-tier infrastructure.

The **object of study** is short-term PM_{2.5} concentration dynamics in Almaty. The **subject of study** is machine-learning models for hourly PM_{2.5} forecasting on low-cost-sensor observations under inversion-dominated mountain-basin conditions.

Research methods include data engineering via the OpenAQ v3 and Open-Meteo APIs, supervised learning with gradient-boosted trees (XGBoost) and Ridge regression, temporal 80/20 train-test splitting, error-regime analysis under varying boundary-layer regimes, and comparison against persistence and CAMS global chemistry-transport model baselines.

The **principal results** are as follows. XGBoost achieves $\text{RMSE} = 23.44 \mu\text{g}/\text{m}^3$ and $R^2 = 0.545$ at the 6-hour forecast horizon on a 2,311-hour held-out winter test set (December 2025 – April 2026), outperforming the CAMS global model by 41% in RMSE. Ridge regression outperforms XGBoost at 12 and 24-hour horizons (RMSE 25.85 and 27.24 $\mu\text{g}/\text{m}^3$ respectively), establishing that model complexity must be matched to forecast horizon and training-set size. All locally trained models substantially outperform CAMS (negative R^2 at all horizons), confirming the value of local LCS-based calibration over freely available global chemistry-transport output.

Scientific novelty: (1) the first publicly evaluated PM_{2.5} forecasting model calibrated for Almaty's LCS network and inversion-dominated winter meteorology; (2) a fully reproducible open-source pipeline directly transferable to other Central Asian cities sharing Almaty's data-availability profile. **Practical significance:** a free-access Streamlit web application providing 6, 12, and 24-hour PM_{2.5} forecasts for Almaty with no authentication requirement.

The thesis consists of **60 pages, 3 figures, 8 tables, 25 sources, 1 appendix.**

Keywords: PM_{2.5} forecasting, Almaty, XGBoost, low-cost sensors, boundary

0.1.2. АНДАТПА (Қазақша)

Экологиялық мәселелерді зерттеу және шешуге арналған интеллектуалды жүйені әзірлеу

Магистрлік диссертация - 7М06106 «Бағдарламалық жасақтама инженериясы» мамандығы, ҚБТУ, 2026 ж.

Ғылыми жетекші: [Аты-жөні, ғылыми дәрежесі]

Жұмыстың **өзектілігі** Қазақстанның Алматы қаласында қысқы мезгілдегі PM2.5 бөлшектерімен ауа ластануының өткір қоғамдық денсаулық мәселесімен анықталады. Қаланың тау қазаншұңқыры топографиясы мен жеке сектордың көмірмен жылытуы инверсия эпизодтарын туғызады: атмосфера шекаралық қабатының биіктігі 10–50 м-ге дейін төмендейді, ал PM2.5 тәуліктік орташа концентрациясы 74–81 мкг/м³-ге жетеді - бұл ДДҰ нормасынан (15 мкг/м³) бес-алты есе жоғары. Тұрғындар мен қалалық қызметтер үшін ешқандай ашық, қайта шығарылатын, жергілікті калибрленген қысқамерзімді болжам болмаған.

Жұмыстың **мақсаты** - тек ашық деректер көздеріне негізделген және тегін бұлттық инфрақұрылымда орналастырылатын Алматы үшін PM2.5 концентрациясы қысқамерзімді болжайтын интеллектуалды жүйені әзірлеу және бағалау.

Зерттеу объектісі - Алматыдағы PM2.5 концентрациясының қысқамерзімді динамикасы. **Зерттеу пәні** - инверсиялы тау котловинасы жағдайында арзан датчиктер (LCS) деректері негізінде PM2.5 сағаттық болжауына арналған машиналық оқыту модельдері.

Зерттеу әдістері: OpenAQ v3 және Open-Meteo API арқылы деректер инженериясы, XGBoost градиенттік бустинг және жоташа регрессия негізінде бақылаулы оқыту; 80/20 уақыттық бөлу бойынша тексеру; қателік режимдерін талдау; персистентті болжам мен CAMS жаһандық химиялық тасымал моделімен салыстыру.

Негізгі нәтижелер: XGBoost 6 сағаттық горизонтта RMSE = 23,44 мкг/м³ және $R^2 = 0,545$ мәндерін көрсетеді, CAMS-тан RMSE бойынша 41%-ға асып озады. Жоташа регрессия 12 және 24 сағаттық горизонттарда XGBoost-тан жақсырақ нәтиже береді (RMSE тиісінше 25,85 және 27,24 мкг/м³). Барлық жергілікті үйренілген модельдер барлық горизонттарда CAMS-тан едәуір жоғары нәтиже береді ($R^2 \leq -0,18$).

Ғылыми жаңалығы: (1) Алматының LCS желісі мен инверсиялы қыс метеорологиясына калибрленген бірінші жарияланған PM2.5 болжау моделі; (2) Орталық Азияның басқа қалаларына ауыстырылатын толық шығарылатын ашық бағдарламалық желі. **Практикалық маңыздылығы:** аутентификация қажет етпейтін, Алматы үшін 6, 12 және 24 сағаттық PM2.5 болжамдарын ұсынатын ашық Streamlit веб-қосымшасы.

Диссертация **60 беттен, 3 суреттен, 8 кестеден, 25 дереккөзден, 1 қосымшадан** тұрады.

Түйін сөздер: PM2.5 болжау, Алматы, XGBoost, арзан датчиктер, шекаралық кабат биіктігі, ауа сапасы, градиенттік бустинг

0.1.3. АННОТАЦИЯ (Русский)

Разработка интеллектуальной системы для исследования и решения экологических проблем

Магистерская диссертация - специальность 7M06106 «Программная инженерия» КБТУ, 2026 г.

Научный руководитель: [ФИО, учёная степень]

Актуальность работы определяется острой проблемой загрязнения атмосферы воздуха взвешенными частицами PM2.5 в зимний период в Алматы. Горно-котловинная топография города в сочетании с угольным отоплением частного сектора формирует инверсионные эпизоды, при которых высота пограничного слоя атмосферы снижается до 10–50 м, а суточная средняя концентрация PM2.5 достигает 74–81 мкг/м³ - в пять-шесть раз выше норматива ВОЗ (15 мкг/м³). При этом ни одного открытого, воспроизводимого, локально калиброванного краткосрочного прогноза качества воздуха для жителей и городских служб Алматы не существовало.

Цель работы - разработка и оценка интеллектуальной системы краткосрочного прогнозирования концентрации PM2.5 для Алматы на основе исключительно открытых источников данных с развёртыванием на бесплатной облачной инфраструктуре.

Объект исследования - динамика краткосрочных концентраций PM2.5 в атмосфере Алматы. **Предмет исследования** - модели машинного обучения для почасового прогнозирования PM2.5 по данным малозатратных датчиков (LCS) в условиях инверсионной горно-котловинной метеорологии.

Методы исследования: инжиниринг данных через API OpenAQ v3 и Open-Meteo; обучение с учителем на основе градиентного бустинга (XGBoost) и гребневой регрессии; валидация с временным разбиением 80/20; анализ режимов ошибок при различной высоте пограничного слоя; сравнение с моделями персистентности и глобальной химико-транспортной моделью CAMS.

Основные результаты: XGBoost достигает RMSE = 23,44 мкг/м³ и R² = 0,545 на горизонте 6 часов на отложенной тестовой выборке из 2 311 часов (декабрь 2025 - апрель 2026), превосходя CAMS на 41% по RMSE. Гребневая регрессия превосходит XGBoost на горизонтах 12 и 24 часа (RMSE 25,85 и 27,24 мкг/м³ соответственно). Все локально обученные модели существенно превосходят CAMS (R² ≤ -0,18 на всех горизонтах), подтверждая ценность локальной калибровки на LCS-данных.

Научная новизна: (1) первая опубликованная модель прогнозирования PM2.5, откалиброванная для сети LCS Алматы и инверсионной зимней метеорологии; (2) полностью воспроизводимый открытый программный конвейер, переносимый на другие города Центральной Азии. **Практическая значимость:** веб-приложение

Streamlit с открытым доступом, обеспечивающее прогнозы PM2.5 на горизонтах 6, 12 и 24 часа для Алматы без необходимости аутентификации.

Диссертация состоит из **60 страниц, 3 рисунков, 8 таблиц, 25 источников, 1 приложения.**

Ключевые слова: прогнозирование PM2.5, Алматы, XGBoost, малозатратные датчики, высота пограничного слоя, качество воздуха, градиентный бустинг

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Introduction

0.1.4. Motivation and relevance

Almaty's winter air pollution is one of the worst urban public-health problems in Central Asia. The city sits in a mountain basin at 700–900 m above sea level, enclosed to the south by the Zailiyskiy Alatau range and open to the steppe to the north. Under anticyclonic winter conditions, a stable temperature inversion forms over the basin, suppressing vertical mixing and trapping combustion products near the surface. During the worst episodes, the atmospheric boundary-layer height (BLH) falls from a summer mean of around 1,000 m to 10–50 m, cutting the effective dilution volume by a factor of 20 to 100. Coal combustion is the dominant emission source in these episodes: most residential heating in peri-urban Almaty runs on solid-fuel boilers, which emit PM_{2.5} and black carbon at rates that depend strongly on combustion temperature and coal quality. Near-zero ventilation combined with high emission rates drives daily-mean PM_{2.5} above 100 $\mu\text{g}/\text{m}^3$ at peak winter, putting the city among the most polluted in the IQAir annual rankings for several consecutive years (IQAir, 2025). The WHO 24-hour PM_{2.5} guideline is 15 $\mu\text{g}/\text{m}^3$ (World Health Organization, 2021); Almaty exceeds it by a factor of six or more during inversion events. Monthly means of 80.7 $\mu\text{g}/\text{m}^3$ (December 2025) and 74.2 $\mu\text{g}/\text{m}^3$ (January 2026) were recorded in the sensor network used in this study.

Until roughly 2023, the public data infrastructure for monitoring this problem was thin. Almaty's official Kazhydromet network ran fewer than 10 reference-grade stations across a 682 km² urban area, too sparse to resolve the fine-scale spatial gradients that basin-trapped pollution produces. The rise of community-scale low-cost sensor (LCS) networks changed this. The AirGradient open-hardware platform, combined with the OpenAQ data aggregation API (OpenAQ, 2024), brought a rapid spread of low-cost monitors across the city. As of April 2026, the network used in this study had 192 sensors within a 25 km radius of the city centre (43.24°N, 76.89°E), of which 153 were AirGradient units concentrated in central and southern residential districts. At roughly one sensor per 3.5 km² on average, with higher density in the most populated zones, this network gives a city-scale spatial picture that the reference network alone could not provide. Other Central Asian cities with similar inversion-driven pollution problems, including Shymkent (Kazakhstan), Karaganda (Kazakhstan), Bishkek (Kyrgyzstan), and Tashkent (Uzbekistan), share the same broad data profile: sparse official monitoring, a growing LCS presence on OpenAQ, and no published machine-learning forecasting work tailored to their geography.

Short-term PM_{2.5} forecasting is well developed in European and North American settings, where reference-grade training targets and dense meteorological networks are available. Gradient-boosted tree models and long short-term memory networks trained on ground-based or satellite-derived features routinely achieve useful skill at 6–24 h horizons. Almaty falls outside this literature for two reasons. First, data access: the global platforms that publish PM_{2.5} forecasts, IQAir, AirVisual, and

BreezoMeter, run proprietary pipelines whose architectures, training data, and evaluation protocols are not public. Second, institutional capacity: Kazhydromet publishes hourly observations at its stations but has no public forecast API or machine-readable hazard thresholds. No peer-reviewed study has trained a gradient-boosted model on Almaty’s LCS network and evaluated it against a freely available calibrated baseline on a held-out winter test period. That gap is the direct motivation for this work.

This work matters beyond Almaty. Central Asia carries one of the highest per-capita PM2.5 disease burdens globally, driven by arid dust sources, coal-heavy energy systems, and fast-growing cities. A reproducible pipeline that needs only an OpenAQ API key, a free Open-Meteo weather archive endpoint (Zippenfenig, 2023), and standard Python libraries is directly portable to any LCS-served city in the region. City centre coordinates, API query windows, and training configuration all live in a single configuration file, so researchers or municipal agencies in Bishkek, Shymkent, or Tashkent could adapt the system to their local network in hours, with no proprietary data or institutional data-sharing agreements required.

0.1.5. Problem statement

There is no open, reproducible short-term PM2.5 forecasting system built on Almaty’s LCS network, and no published evaluation of how a locally trained model compares to the freely available CAMS global forecast for this geography. The core scientific question is how much predictive skill is gained by training a machine-learning model on Almaty’s own sensor data versus applying an off-the-shelf global chemistry-transport model.

Two subsidiary questions follow. First, given the limited LCS dataset (roughly 9,242 training hours across 562 days), it is not clear whether a non-linear gradient-boosted model or a regularised linear model gives better accuracy; the answer has direct implications for model selection in data-scarce cities elsewhere in Central Asia. Second, the spatial and temporal representativeness of a city-median derived from an unevenly distributed LCS network has not been quantified for Almaty. During outage periods the median may reflect fewer than 30 active stations rather than the nominal 192, and what that means for forecast validity is unknown. Answering these questions requires a complete evaluation pipeline running on real, openly accessible data from Almaty’s sensor network, not on synthetic or reference-grade surrogates. This study provides that pipeline and reports the first quantitative answers to both questions over a 562-day window ending April 2026.

0.1.6. Aim and objectives

Aim. Develop and evaluate an intelligent short-term PM2.5 forecasting system for Almaty that is built entirely on open data and code and is deployable on free-tier infrastructure.

Objectives. 1. Ingest and curate PM2.5 observations (OpenAQ LCS) and meteorological features (Open-Meteo, including boundary-layer height) for the 2024-10-01 → 2026-04-15 window into a reproducible SQLite store. 2. Engineer temporal and meteorological features tailored to Almaty’s winter inversion regime. 3.

Train and cross-validate an XGBoost forecaster for 6/12/24 h horizons and compare it against three baselines: persistence, linear regression, and CAMS. 4. Analyse error regimes - in particular, episodes of sudden onset and rapid clearing - and articulate the model's failure modes. 5. Ship a single-screen Streamlit web application that visualises current observations and the XGBoost forecast on a map of Almaty.

0.1.7. Scientific novelty

This study makes three contributions. The first is the initial publicly released PM2.5 forecasting model calibrated for Almaty's LCS network and inversion-dominated winter meteorology. Prior work on Almaty air quality is mostly descriptive (source apportionment, trend analysis) or relies on satellite-derived estimates rather than ground-level sensor data. The model here is trained and evaluated on the 192-station OpenAQ network over a 562-day window ending April 2026, using a feature set that encodes BLH dynamics and inversion-onset signals absent from earlier local work.

The second contribution is the reproducible open architecture. All inputs are openly licensed (CC BY 4.0, no paywalled data); the full pipeline is a single Python repository with no cloud dependencies. Comparable published systems, including those evaluated by (Ayus et al., 2023)], typically rely on proprietary ground truth, restricted API access, or institutional data agreements that block reproduction or geographic transfer. This design removes those barriers, making the pipeline portable to Shymkent, Karaganda, and other Central Asian cities with analogous LCS-on-OpenAQ data profiles.

The third contribution is a concrete result on model-complexity trade-offs. Ridge regression outperforms XGBoost at the 12 and 24-hour horizons: by 9.8% in RMSE at +12 h (Ridge: 25.85 vs. XGBoost: 28.67 $\mu\text{g}/\text{m}^3$) and by 13.5% at +24 h (Ridge: 27.24 vs. XGBoost: 31.48 $\mu\text{g}/\text{m}^3$). This result comes from a training set of 9,242 hours with 27 features, a scale at which non-linear ensemble methods would normally be expected to retain their advantage based on published benchmarks. The finding adds to the evidence on model-complexity trade-offs for air quality forecasting with under 10,000 training samples. The most closely related published comparisons (Ayus et al., 2023; Mahajan et al., 2018)] report similar reversals in settings where the target variable carries substantial measurement noise, consistent with the known hygroscopic overestimation bias in uncalibrated LCS readings.

The reversal also has a direct consequence for system design. Ridge regression needs no tree construction, so its inference time on a single feature row is negligible compared to XGBoost, and its serialised model file is orders of magnitude smaller. On a free-tier Streamlit deployment where memory and compute are tight, a Ridge model at the 12 and 24-hour horizons is not just equally accurate but the better engineering choice. That alignment of statistical and operational considerations makes the case for a multi-model system rather than a single one.

0.1.8. Object, subject, and methods

- **Object of study:** short-term PM2.5 concentration dynamics in Almaty.
- **Subject of study:** machine-learning models for hourly PM2.5 forecasting on

LCS-based observations in inversion-dominated mountain-basin conditions.

- **Methods:** data engineering on openly available APIs; supervised learning with gradient-boosted trees; time-series cross-validation; baseline comparison; error-regime analysis.

0.1.9. Practical significance

The locally trained XGBoost model achieves $\text{RMSE} = 23.44 \mu\text{g}/\text{m}^3$ and $R^2 = 0.545$ at the 6-hour forecast horizon on a held-out winter test set of 2,311 hours (December 2025 – April 2026), outperforming the CAMS global chemistry-transport model by 41.1% in RMSE at the same horizon. At the 12-hour horizon, Ridge regression produces $\text{RMSE} = 25.85 \mu\text{g}/\text{m}^3$ and $R^2 = 0.446$, outperforming CAMS by approximately 28% in RMSE. At 24 hours, Ridge achieves $\text{RMSE} = 27.24 \mu\text{g}/\text{m}^3$ and $R^2 = 0.383$, while CAMS produces negative R^2 values at all three horizons (-0.314 , -0.376 , and -0.184 at 6, 12, and 24 h respectively), indicating that the global model provides no useful predictive signal for this geography. All locally trained models outperform CAMS by at least 16% in RMSE across every horizon tested.

These results are deployed as a Streamlit web application that provides a continuously updated PM_{2.5} forecast for Almaty at no cost. The application refreshes every 30–60 seconds, requires no authentication, and runs in any browser, making it usable by schools, hospitals, municipal public health offices, and community organisations without technical staff or data infrastructure. Forecast uncertainty is shown as a confidence band of roughly $\pm 17 \mu\text{g}/\text{m}^3$ at the 6-hour horizon (± 1 MAE). An advisory triggered when the lower bound crosses the WHO 24-hour guideline of $15 \mu\text{g}/\text{m}^3$ will catch genuine deterioration events at a false-positive rate acceptable for public communication. The map view shows the current city-median PM_{2.5} alongside the three horizon forecasts; the colour scale follows Kazakhstani national AQI categories so non-specialist users can read the values without knowing $\mu\text{g}/\text{m}^3$ numerics. The full pipeline, covering data ingest, feature engineering, model training, and the application, is a single open Python repository with no cloud dependencies and is directly portable to other Central Asian cities.

Deployment URL: [to be added after Week 3 deployment]

0.1.10. Structure of the thesis

Chapter 1 reviews the scientific and policy context: the meteorological and emission drivers of Almaty’s PM_{2.5} episodes (temperature inversions, coal combustion); the low-cost sensor technology and the OpenAQ data ecosystem behind this dataset; and the methodological literature on gradient-boosted tree models for short-term urban PM_{2.5} forecasting, with emphasis on inversion-dominated mountain or basin environments. Chapter 2 describes the data sources and system architecture in reproducible detail, covering the SQLite database schema, the OpenAQ and Open-Meteo API ingestion procedures, the 27-feature matrix (BLH delta, temporal lag, and calendar features), the train/test split, and the rationale for each non-obvious design choice. Chapter 3 reports the experimental results: quantitative model comparisons across

all three forecast horizons and all four models, feature importance, error-regime case studies on inversion onset and clearing events, and the spatial analysis of sensor-coverage effects during the October–November 2025 AirGradient firmware outage. Chapter 4 discusses what the results mean for the stated research questions, places the findings within the published literature, quantifies the main limitations, and proposes five directions for future work, including SHAP-based feature attribution, forecast BLH integration, and an XGBoost–Ridge ensemble. An appendix gives the full model hyperparameter configurations, API query specifications, and data-quality filter thresholds applied during ingestion, so the complete pipeline can be reproduced from the archived code and database.

1. Analytical Review

1.0.1. 1.1. Air quality in Almaty and Kazakhstan

1.0.1.1 1.1.1. Topographic and meteorological drivers

Almaty occupies a basin at the northern foothills of the Trans-Ili Alatau mountain range (43.24°N, 76.89°E, elevation 700–900 m above sea level), with peaks rising to over 4,000 m immediately to the south and south-east. This asymmetric terrain configuration is the primary determinant of the city’s air quality regime, and it operates through two interacting mechanisms: cold-air drainage and thermally forced boundary-layer collapse.

Cold-air drainage occurs because the dense, chilled air that accumulates overnight on the mountain slopes flows downslope by gravity and pools in the Almaty basin. The surrounding terrain to the north, where the city transitions to the Kazakhstani steppe, offers relatively unobstructed outflow, but the valley geometry slows horizontal ventilation considerably. The result is that cold surface-air masses can persist for 12–36 hours after the initial nocturnal drainage event, particularly when synoptic-scale anticyclonic conditions suppress vertical mixing over the entire region. During sustained inversion events recorded in the training dataset (2024-10-01 to 2026-04-15), boundary-layer heights (BLH) fall to as low as 10 m, against a summer mean of approximately 1,000–1,500 m and a training-window mean of 257 m. This extreme range - 10 m at the low end, 3,935 m at the peak of thermally active summer afternoons - is a quantitative signature of the inversion-dominated climate and is directly exploited by the BLH feature in the forecasting model.

Temperature inversions arise when the vertical temperature profile of the atmosphere is inverted relative to the standard lapse rate: the air aloft is warmer than the surface layer. Under such conditions, vertical convective mixing is suppressed because a parcel of surface air, if displaced upward, encounters air that is warmer and thus more buoyant, and it sinks back. In Almaty, radiation inversions develop on clear winter nights when the snow-covered surface loses heat rapidly through long-wave emission, chilling the near-surface air. Surface-based inversions of 5–15 °C have been documented in regional meteorological records (Kerimray et al., 2020; Tursumbayeva et al., 2022)]. Elevated inversions also form when warm air advects from the south over the mountain range and overrides the cold pool in the basin. Both inversion types act as a lid that confines surface-emitted aerosols to the lower 50–200 m of the atmosphere, where the city’s population lives and breathes.

The seasonal modulation of boundary-layer depth explains why Almaty’s PM_{2.5} problem is almost entirely a winter phenomenon. In summer, strong solar heating destroys the nocturnal inversion by mid-morning and drives the convective boundary layer to 1,500–3,000 m, diluting surface emissions into a volume that is ten to twenty times larger than the winter surface layer. This dilution effect, independent of changes in emissions, is sufficient to reduce ambient PM_{2.5} concentrations from winter highs exceeding 100 µg/m³ to the 12–17 µg/m³ summer baseline observed

in the present dataset. The seasonality of PM_{2.5} in Almaty is therefore not a pure reflection of emission seasonality; it is the product of emission intensity multiplied by a highly variable dilution volume.

1.0.1.2 1.1.2. Emission sources and seasonal patterns

Almaty’s PM_{2.5} comes from three sources: residential coal combustion, thermal power generation, and vehicle traffic. Their relative contributions shift with season and time of day, and their combined effect is amplified by the meteorological drivers described in §1.1.1.

Residential coal heating is the dominant wintertime source. Peri-urban and low-income districts on the city periphery, particularly in the eastern and western micro-districts, lack connection to the centralised district heating network. Households in these areas rely on individual coal-fired stoves or boilers, which operate at much lower combustion temperatures than utility-scale furnaces and therefore emit disproportionately high quantities of unburned carbonaceous particles, fine particulate matter, and black carbon. The heating season in Kazakhstan officially runs from approximately 15 October to 31 March, corresponding to the cold-temperature window when residential heating demand is greatest. This six-month window almost exactly coincides with the high-PM_{2.5} months in the dataset: the monthly mean of 80.7 $\mu\text{g}/\text{m}^3$ in December 2025 and 74.2 $\mu\text{g}/\text{m}^3$ in January 2026, against a summer baseline of 12–17 $\mu\text{g}/\text{m}^3$, define a seasonal amplitude factor of approximately five. The sharpness of the heating-season transition - visible in the training time series as a rapid concentration increase in mid-October - motivated the inclusion of a binary heating-season indicator (1 from 15 October to 31 March, 0 otherwise) as a categorical feature in the model.

Thermal power generation contributes through Almaty’s two combined-heat-and-power plants, CHP-2 (located in the city’s western industrial zone) and CHP-3 (in the northern periphery). Both plants burn Karaganda-basin coal and emit PM_{2.5}, sulfur dioxide, and nitrogen oxides from stacks whose effective injection height exceeds the inversion-layer ceiling under calm conditions. During inversion events, emissions that would normally disperse into the mixed layer instead accumulate near-surface after dry deposition and secondary aerosol formation. The plants collectively supply approximately 70% of the city’s district heating load (Kerimray et al., 2020)], making them non-dispatchable during the heating season.

Vehicle traffic compounds the problem through a different diurnal cycle. Almaty’s passenger car ownership rate is among the highest in Central Asia, and a substantial fraction of the fleet is older petrol and diesel vehicles with limited emissions control. Traffic-related PM_{2.5} - primarily brake dust, tyre wear particles, and direct exhaust - peaks during morning (07:00–09:00 local time) and evening (17:00–19:00) commutes. This diurnal pattern overlaps with the weakest boundary-layer mixing of the day, creating a double-peaked hourly concentration signature that is reproduced in the lag feature structure of the model. The hour-of-day cyclical encoding captures

this pattern without requiring explicit source attribution.

A secondary emission category relevant to summertime periods is dust resuspension. The Almaty basin is bordered to the north by semi-arid steppe, and dry southerly winds in spring carry mineral dust from bare soil surfaces. These dust events can briefly elevate PM_{2.5} to 40–60 µg/m³ even in months outside the heating season, producing short-duration spikes that are visible as outliers in the summer baseline series. Dust events are not the focus of the present model, but they introduce heteroscedastic residuals in the spring transition period (April–May) that affect the RMSE decomposition in §3.4.

1.0.1.3 1.1.3. Health impacts and regulatory context

Sustained PM_{2.5} exposure at Almaty’s observed concentrations has severe health consequences. The World Health Organization’s 2021 air quality guidelines set the 24-hour PM_{2.5} limit at 15 µg/m³ and the annual mean guideline at 5 µg/m³ (World Health Organization, 2021). During the heating season of 2025–26, the cross-sensor median PM_{2.5} in the present dataset exceeded the 24-hour guideline on the majority of winter days. At exposures above 35 µg/m³, the WHO epidemiological evidence base links each 10 µg/m³ increment in 24-hour PM_{2.5} to increases in all-cause mortality, cardiovascular hospital admissions, and respiratory emergency department visits (World Health Organization Regional Office for Europe, 2013)]. Almaty’s approximately two million residents are exposed to these concentrations for roughly six months of each year.

Kazakhstan’s regulatory framework for ambient air quality is established by the Environmental Code of the Republic of Kazakhstan, enacted in January 2021 (Экологический кодекс Республики Казахстан, 2021). The Code sets maximum permissible concentrations (MPC) for PM_{2.5}, mandates environmental impact assessment for major emission sources, and establishes a framework for emission trading and environmental permits. The national AI strategy for 2021–2025 identifies environmental monitoring and smart city applications as priority domains for artificial intelligence deployment. Despite this policy context, there is no publicly accessible, near-real-time, locally calibrated short-term PM_{2.5} forecast for Almaty that is openly documented or independently reproducible. This work directly addresses this gap.

The gap between regulatory intent and operational reality has several causes. Kazhydromet’s reference monitoring network, while technically credible, is too sparse to provide the neighbourhood-level spatial detail needed for public exposure guidance, and its data publication lag makes real-time decision support impossible (§1.2.1). The commercial forecast services that do exist for Almaty (§1.2.3) operate without published evaluation metrics. Public health interventions - school closures on high-pollution days, voluntary traffic restrictions, biomass burning bans - require actionable 6–24 hour forecasts that are not currently available from an open, auditable source.

1.0.2. 1.2. Existing monitoring systems

1.0.2.1 1.2.1. Reference-grade monitoring

The formal air quality monitoring network in Almaty is operated by Kazhydromet, the national hydrometeorological and environmental monitoring service of the Republic of Kazakhstan. Reference-grade monitoring stations in this network use β -attenuation monitors (BAM) or tapered-element oscillating microbalance (TEOM) instruments that meet World Meteorological Organization (WMO) performance requirements for PM_{2.5} mass concentration measurement and provide readings traceable to gravimetric standards. Both instrument types are widely accepted internationally as reference-equivalent methods: BAM instruments operate by drawing a particle-laden sample through a filter tape and measuring the attenuation of a beta-radiation beam, while TEOM instruments determine mass accumulation from the resonance frequency shift of a vibrating glass element. Both achieve measurement uncertainties of ± 5 –10% under typical operating conditions (World Meteorological Organization, 2016)].

The Kazhydromet network is sparse for a city of Almaty's scale. Fewer than ten operational stations are distributed across a city area of approximately 682 km², yielding an average area per station above 68 km². For comparison, the US EPA NAAQS monitoring guidance recommends station spacing on the order of 10–20 km in urban areas with known emission hotspots, while population-exposure studies typically require even finer resolution to capture neighbourhood-scale gradients. At the observed station density, the Kazhydromet network cannot resolve intra-urban spatial gradients, identify emission hotspots, or provide the spatially aggregated city-scale concentration that would serve as a single-number target for a forecasting model.

Equally constraining for the purposes of this work is the data accessibility limitation. Kazhydromet does not expose a public, machine-readable API for near-real-time hourly PM_{2.5} data. Data are published in periodic bulletins and annual reports, with lags of days to months between measurement and public release. This lag structure makes Kazhydromet data unsuitable as a training target or real-time forecast anchor for an hourly prediction system. This study therefore does not use Kazhydromet data as a primary input, though a future co-location study comparing the OpenAQ LCS median against Kazhydromet reference readings would substantially strengthen the quantitative interpretation of model results.

1.0.2.2 1.2.2. Low-cost sensor networks

The community low-cost sensor (LCS) network aggregated through the OpenAQ v3 platform substantially densifies Almaty's monitoring landscape (OpenAQ, 2024). As of the API snapshot taken on 20 April 2026, 192 active monitoring locations within 25 km of the city centre (43.24°N, 76.89°E) report PM_{2.5} data through the OpenAQ platform, providing a spatial density roughly twenty times greater than the Kazhydromet reference network.

The dominant contributor to this network is the Almaty Air Initiative, a citizen-science organisation that has deployed AirGradient ONE open-hardware monitors across residential districts since 2024. AirGradient units account for 153 of the 192 stations (79.7%). AirGradient ONE is an open-source, community-built sensor platform based on the Plantower PMS5003 optical particle counter, whose design schematics and firmware are publicly available under an open-hardware licence. The Clarity Node-S, a commercial LCS device manufactured by Clarity Movement Co. (Berkeley, CA), contributes 38 stations (19.8%). A single AirNow-affiliated station provides the remaining 0.5% of the network. All data transmitted through OpenAQ are licenced under CC BY 4.0, permitting unrestricted research use with attribution. This open-licence condition is a prerequisite for the reproducibility objective of this work.

The OpenAQ platform’s v3 API provides a standardised JSON interface through which sensor readings, metadata, and quality flags can be queried programmatically. The ingest pipeline described in §2.2 issues hourly requests to the `/v3/locations/{id}` endpoint, filters readings by the plausibility criteria defined in §2.3, and assembles the city-wide PM2.5 median used as the model target. The combination of platform standardisation and open licensing makes OpenAQ the only viable data source for a fully reproducible Almaty forecasting pipeline as of the study period.

1.0.2.3 1.2.3. Commercial aggregators and the data gap

Commercial air quality platforms - notably IQAir and the AirKz aggregator - publish AQI dashboards and, in some cases, AI-based short-term forecasts for Almaty. IQAir’s World Air Quality Report (IQAir, 2024) ranked Almaty among the most polluted cities in Central Asia, citing annual mean PM2.5 concentrations that are consistent with the seasonal patterns documented in the present dataset. Both platforms play a useful role in public communication.

Their limitation for scientific purposes is fundamental: neither IQAir nor AirKz publishes methodology documentation, calibration procedures, model architecture descriptions, evaluation metrics, or the underlying training data for their forecasting products. The IQAir forecast pipeline, in particular, ingests data from a global network of IQAir AirVisual Pro sensors and applies undisclosed correction algorithms whose performance has not been independently peer-reviewed. Without access to methodology, forecast skill cannot be independently assessed. Without access to training data, the published rankings cannot be reproduced. Without published evaluation against held-out data, any claimed accuracy is unverifiable.

This opacity creates a specific gap that this work addresses: Almaty lacks a published, open, and reproducible short-term PM2.5 forecast whose inputs, processing steps, model architecture, and evaluation results are fully documented. The present system achieves a 41% RMSE improvement over the CAMS global baseline at the 6-hour horizon (§3.2), and every step of the pipeline, from API ingest to Streamlit deployment, is version-controlled and replicable by any researcher with access to the

public OpenAQ and Open-Meteo APIs.

The reproducibility requirement is not merely an academic formality. A forecast system that informs public health decisions must be auditable: a regulator, a public health official, or a peer reviewer must be able to trace any individual forecast back to its training data, feature specification, and model parameters. Closed commercial pipelines cannot meet this standard. This work addresses it by publishing the complete pipeline source code, the BibTeX-documented data provenance, and a held-out test set evaluation on 2,311 hours of winter 2025–26 data that was excluded from all training and hyperparameter selection steps. Any user of the open-source repository can reproduce the $\text{RMSE} = 23.44 \mu\text{g}/\text{m}^3$ result at the 6-hour horizon from scratch using only freely available data and a standard Python environment.

1.0.3. 1.3. Low-cost sensors: capabilities and limitations

1.0.3.1 1.3.1. Operating principles and cost structure

Low-cost sensors (LCS) for PM_{2.5} measurement exploit optical particle counting through laser-scattering. An aerosol sample is drawn through a sensing chamber where a laser diode - typically operating at 780 nm wavelength - illuminates the particle stream. Individual particles or clusters scatter the laser light at angles and intensities that depend on their size and refractive index; a photodetector positioned at a defined scattering angle converts this optical signal to a count distribution across particle-size bins. The sensor firmware then applies a lognormal conversion model to estimate PM_{2.5} mass concentration from the particle-count data.

The two dominant LCS platforms in the Almaty network exemplify the diversity within this operating principle. The Plantower PMS5003, embedded in the AirGradient ONE, draws a sample flow of approximately 100 mL/min through a 34 mm × 20 mm sensing volume and outputs concentration estimates for PM_{1.0}, PM_{2.5}, and PM₁₀ at one-minute intervals via serial UART protocol. The Sensirion SPS30, an alternative optical counter used in some AirGradient ONE units, adds a laser-particle-interaction cleaning cycle that reduces particle deposition on the sensing optics, extending the calibrated service life from approximately 8,000 to 12,000 operating hours (Sensirion AG, 2023)]. The Clarity Node-S employs a proprietary laser-scattering geometry with firmware-level humidity correction, though the correction algorithm is not publicly documented.

The economic case for deploying LCS networks rather than reference instruments is straightforward at these cost levels. Reference-grade BAM and TEOM instruments cost USD 20,000–80,000 per unit, plus installation, calibration, and maintenance costs that can approach the capital cost over a five-year operational period. In contrast, an AirGradient ONE sensor - including enclosure, microcontroller, and connectivity - costs approximately USD 100–150 in component form, and the Clarity Node-S retails for approximately USD 300–400. This 100–500× cost differential makes the 192-station Almaty network economically feasible for a citizen-science organisation, whereas a reference-grade network of comparable density would cost tens of millions of dollars. The trade-off is that LCS measurements lack the metro-

logical traceability of reference instruments and are subject to the biases described in §1.3.2.

1.0.3.2 1.3.2. Humidity and drift biases

The primary known systematic bias for optical LCS in humid environments is hygroscopic growth of aerosol particles. Above approximately 70% relative humidity (RH), water molecules condense onto the surfaces of hygroscopic aerosol constituents - primarily sulfate, nitrate, and organic carbon species - enlarging the apparent particle diameter. Because LCS firmware translates particle diameter to mass concentration using refractive indices and density assumptions calibrated for dry particles, the measured PM_{2.5} is systematically inflated under humid conditions. Published intercomparisons of Plantower-class sensors against reference BAM instruments in urban environments report positive humidity biases of 20–60% at RH > 80%, with the magnitude depending on the hygroscopic fraction of the local aerosol (Di Antonio et al., 2018; Kosmopoulos et al., 2020)]. During Almaty’s winter inversion episodes - precisely the high-concentration events of greatest interest for forecasting - ambient RH frequently exceeds 75–85%, meaning that the LCS readings are likely to be positively biased during the most critical measurement periods.

Inter-unit variability is a second quantifiable source of uncertainty. Factory-calibrated units of the same model show concentration offsets of 10–30% under identical controlled-chamber conditions, attributable to tolerances in laser alignment, photodetector response curves, and firmware parameter sets (Popoola et al., 2018)]. When units are deployed across a city with diverse microclimates, varying heights above ground (some AirGradient units are mounted at 1.5 m, others at 3–5 m on balconies or rooftops), and different local source influences, the inter-unit spread across the 192 Almaty stations during any given hour substantially exceeds the controlled-chamber 10–30% figure. The standard deviation across all active stations in the present dataset can exceed 30 $\mu\text{g}/\text{m}^3$ in a single hour during peak winter episodes, confirming that individual-station readings are not representative of city-wide exposure.

A third bias source is temporal drift: the gradual degradation of laser intensity and photodetector sensitivity over months of field deployment. Laboratory studies of Plantower PMS5003 units subjected to accelerated aging report signal attenuation of 10–25% over approximately 8,000–10,000 operating hours (Searle et al., 2023)]. The training window in this study spans 562 days (2024-10-01 to 2026-04-15), or roughly 13,500 operating hours for sensors that were already deployed before the training window opened. Systematic drift cannot therefore be excluded, particularly for AirGradient ONE units deployed in the inaugural 2024 network expansion. Without periodic reference co-location, the magnitude of drift in the Almaty network is not directly measurable from the available data.

No co-location calibration against reference instruments was performed in this study. The PM_{2.5} values derived from the OpenAQ network should therefore be

understood as relative concentration signals that track true PM_{2.5} variability but may carry systematic biases of $\pm 20\text{--}50\%$ in absolute magnitude during high-RH winter conditions. This limitation is explicitly acknowledged in the scope discussion (§4.2) and does not invalidate the relative accuracy comparisons among models trained and evaluated on the same LCS-derived target series.

1.0.3.3 1.3.3. Mitigation strategies applied in this study

The standard mitigation for inter-unit variability in a multi-sensor LCS network is spatial aggregation: combining readings from multiple co-located or nearby sensors so that random unit-specific offsets cancel in the aggregate. This study applies this principle at city scale by computing the cross-sectional median PM_{2.5} across all stations with plausible readings in a given hour. The median, rather than the mean, is chosen because the distribution of station readings in any given hour is typically right-skewed - a small number of stations near localised sources (a coal stove, a road intersection) read substantially above the bulk of the network, and those outliers would inflate the mean disproportionately. The median is resistant to such outliers by construction.

Three additional constraints bound the aggregation. A minimum of three contributing stations per hour is required; hours with fewer than three readings are recorded as missing rather than propagated as unreliable estimates. A plausibility filter excludes readings outside the range $0\text{--}500\text{ }\mu\text{g}/\text{m}^3$, rejecting physically implausible values likely caused by sensor malfunction or communication errors. A further check discards readings from any station whose 24-hour running mean deviates by more than three standard deviations from the network median, flagging potentially faulty or improperly calibrated units. These thresholds were chosen based on the empirical distribution of the raw API data and are consistent with practices reported in multi-city LCS network studies (Morawska et al., 2018)].

The city-median aggregation procedure does not correct for humidity bias, which is common-mode across the network (all sensors experience the same ambient RH) and therefore does not cancel in the median. This means that the model target series is systematically inflated during high-RH winter episodes relative to the true gravimetric PM_{2.5}. As a consequence, the forecast bias statistics reported in §3.2 reflect errors relative to the LCS-derived target, not relative to reference-grade measurements. This distinction is noted wherever comparative claims are made.

The scientific justification for proceeding with LCS data despite these limitations is that the alternative is no hourly, city-scale, open dataset for Almaty at any quality level. Reference monitoring exists but is not openly accessible in real time. Commercial platforms exist but are closed. The LCS network, with its known limitations managed by the aggregation and filtering procedures described above, is the only viable basis for a reproducible, operationally testable forecasting system. This position is consistent with the broader field consensus that LCS networks, used with appropriate quality controls, extend the reach of air quality science into data-sparse regions where reference monitoring cannot be economically deployed (Snyder et al.,

2013)].

1.0.4. 1.4. Air quality forecasting methods

Short-term PM_{2.5} forecasting methods span five broad methodological families, each with distinct data requirements, theoretical assumptions, computational demands, and performance characteristics. The choice is not just about benchmark error; it depends on dataset size, feature availability, deployment constraints, and the physical regime. This section reviews each family and arrives at an explicit rationale in §1.4.6.

1.0.4.1 1.4.1. Statistical and persistence methods

The simplest forecast for PM_{2.5} at horizon h is persistence: the predicted concentration at time $t + h$ is set equal to the observed concentration at time t . Despite its trivial construction, persistence is a surprisingly competitive baseline for air pollutants at short horizons because PM_{2.5} exhibits strong positive autocorrelation at lags of one to six hours, driven by the relatively slow dynamics of boundary-layer mixing and emission accumulation. At the one-hour horizon, persistence routinely achieves R^2 values of 0.85–0.95 in urban sites with continuous emissions (Mahajan et al., 2018)]. At the 24-hour horizon, persistence degrades substantially because inversion onset and clearing events decouple the current concentration from the next-day value.

Autoregressive integrated moving average (ARIMA) models extend persistence by fitting a parametric autocorrelation and moving-average structure to the observed time series. The Box-Jenkins ARIMA(p, d, q) framework allows the inclusion of seasonal differencing terms - an SARIMA extension - which can capture the diurnal periodicity of PM_{2.5}. When meteorological covariates are added as exogenous regressors, the model is termed ARIMAX or, in state-space form, a dynamic regression model. These models have well-understood statistical properties, produce interpretable coefficients, and require minimal computational resources to fit and forecast. Published applications to urban PM_{2.5} forecasting in moderate-pollution cities report RMSE reductions of 15–25% relative to naive persistence at the 24-hour horizon (Dong et al., 2017)].

The fundamental limitation of ARIMA-class models for the Almaty setting is their linearity. The onset of a temperature inversion is a non-linear threshold event: boundary-layer height can collapse from 800 m to 50 m within two to four hours as a function of synoptic pressure pattern, surface energy balance, and cold-air drainage dynamics. The relationship between BLH and PM_{2.5} concentration is strongly non-linear - a halving of BLH roughly doubles surface concentration if emissions are held constant, and the emissions themselves are non-linearly dependent on temperature (colder temperatures increase residential heating demand). ARIMA cannot represent these interaction effects without explicit interaction-term specification that would require accurate physical knowledge of the inversion onset mechanism. Kalman filter methods, which model the concentration as a latent state evolving through a linear Gaussian process, share this linearity constraint, though extended and unscented vari-

ants can accommodate mild non-linearities at the cost of increased tuning complexity.

1.0.4.2 1.4.2. Classical machine learning on tabular features

Gradient-boosted decision tree (GBDT) ensembles and random forests constitute the dominant paradigm in operational short-term urban AQ forecasting as of this study period. These methods work by training an ensemble of decision trees on a tabular feature matrix whose columns include autoregressive lags of PM_{2.5}, meteorological variables, and temporal encodings such as hour-of-day, day-of-week, and heating-season indicators. Each tree in the ensemble learns a non-linear partition of the feature space that explains residual variance not captured by the preceding trees. The gradient-boosting optimisation framework - introduced in the XGBoost formulation by Chen and Guestrin (2016) (Chen and Guestrin, 2016)] and subsequently refined in LightGBM (Ke et al., 2017)] - adds L1 and L2 regularisation terms to the tree-growing objective, which substantially reduces overfitting on moderately sized datasets.

Published comparisons of GBDT-class models against ARIMA and persistence baselines on urban PM_{2.5} data consistently report RMSE improvements of 20–40% at 24-hour horizons (Ayus et al., 2023; Feng et al., 2015)]. The improvement is most pronounced in cities with strong meteorological forcing - cold-season inversions, monsoon transitions, dust events - because the tree-splitting mechanism can capture the non-linear meteorology-concentration interactions that linear models cannot. For the Almaty setting, the inversion-driven concentration regime described in §1.1.1 is precisely the regime where GBDT methods offer the largest gains over statistical alternatives.

A key design decision in GBDT-based forecasting is the construction of the lag feature set. For a forecast horizon of h hours, the model cannot use observations from time $t + 1$ through $t + h$ because these are unknown at forecast time. The available autoregressive inputs are therefore observations at lags 1, 2, ..., L beyond the earliest available observation, where L is a hyperparameter chosen to balance memory depth against feature dimensionality. This study uses lags at 1, 2, 3, 6, 12, and 24 hours for PM_{2.5} and selects meteorological features at forecast-time (using Open-Meteo numerical weather prediction output for the future window) (Zippenfenig, 2023). This lag structure was determined by cross-correlation analysis of the training series and is described in detail in §2.4.

Random forests - the bagged-tree precursor to gradient boosting - are less commonly reported as primary models in recent urban AQ literature, having been largely superseded by XGBoost and LightGBM in competitive benchmarks. They remain valuable as ablation baselines because they share the tree-ensemble architecture with GBDT but use averaging rather than gradient optimisation, making their performance differential against XGBoost a direct measure of the benefit from boosting regularisation.

1.0.4.3 1.4.3. Deep learning approaches

Long short-term memory (LSTM) networks and their gated recurrent unit (GRU) variants extend the sequence modelling capability of standard recurrent neural networks (RNNs) by introducing gating mechanisms - input, forget, and output gates in LSTM; update and reset gates in GRU - that regulate the flow of information through the hidden state over time. These gates allow the network to retain relevant information across sequences of hundreds to thousands of time steps without the vanishing-gradient pathology that limits plain RNNs. Applied to air quality forecasting, LSTM models replace the hand-crafted lag features of GBDT approaches with learned temporal dependencies, potentially capturing multi-scale autocorrelation structures that manual lag selection misses.

Published applications of LSTM to urban PM_{2.5} forecasting report competitive performance at 24–72 hour horizons when training sets exceed approximately 20,000–50,000 time steps (Qi et al., 2019)]. At this scale, the capacity of the LSTM to learn non-linear temporal patterns appears to justify its additional complexity relative to GBDT models. However, across a wider survey of urban AQ forecasting studies, the reported gains of LSTM over gradient-boosting are inconsistent: approximately half of published comparisons find LSTM superior, and half find GBDT competitive or better, with the outcome depending strongly on dataset size, number of monitoring stations, and the degree of feature engineering applied to the GBDT competitor (Lim and Zohren, 2021)]. The inconsistency suggests that much of the reported LSTM advantage in favourable studies arises from GBDT underfitting due to insufficient lag features rather than from a fundamental architectural advantage of recurrent networks.

Convolutional-LSTM (Conv-LSTM) and CNN-LSTM hybrid architectures have been proposed for air quality settings with dense spatial monitoring networks, where the convolutional layers extract spatial concentration patterns and the LSTM layers model their temporal evolution (Ding et al., 2021)]. These methods require the monitoring network to have consistent spatial coverage - ideally a regular grid - because the convolution operation is defined over a spatially regular input. The 192 Almaty LCS stations are distributed unevenly, with high density in central residential districts and sparse coverage in industrial and peri-urban areas. Interpolating these stations onto a regular grid would introduce kriging or bilinear-interpolation artefacts, and the resulting synthetic spatial fields would not carry independent information beyond the city-median series already used as the model target. Spatially explicit deep learning is therefore not applicable to the present dataset without additional investment in network spatial design.

Transformer-based models, following the self-attention mechanism of Vaswani et al. (2017) (Vaswani et al., 2017)], have been adapted for time-series forecasting under the PatchTST, Informer, and Autoformer architectures. These models handle long-range dependencies through attention rather than recurrence and have demonstrated competitive performance on long-horizon (168–720 step) forecasting bench-

marks. For the short-horizon PM_{2.5} task considered here - 6, 12, and 24 hours - the self-attention mechanism’s advantage over GBDT lag features is marginal, because the relevant dependencies are contained within a 24–48 hour window that GBDT can represent compactly with a moderate-length lag vector. Transformer models also require substantially more training data than the approximately 9,200 hourly rows available in this study; typical transformer time-series benchmarks use 10,000–100,000+ training instances per target series to prevent overfitting of the large parameter count.

The deployment context additionally constrains model selection. This study targets a Streamlit application running on a free-tier hosting environment with CPU-only inference. LSTM and transformer models require either a GPU for efficient inference or a model compression step (ONNX export, TensorFlow Lite) that adds pipeline complexity. XGBoost models serialised as 2 MB Joblib files produce forecasts in under 20 milliseconds on a single CPU core, satisfying the latency requirement without compression or GPU acceleration.

1.0.4.4 1.4.4. Physics-informed and hybrid methods

Chemistry-transport models (CTMs) simulate atmospheric concentrations by solving the advection-diffusion-reaction equations governing the transport, chemical transformation, and deposition of pollutants in a discretised atmospheric domain. The Copernicus Atmosphere Monitoring Service (CAMS), operated by the European Centre for Medium-Range Weather Forecasts (ECMWF), provides freely accessible global hourly PM_{2.5} analyses and forecasts at approximately 40 km horizontal resolution and 60 vertical levels [Inness et al., 2019]. WRF-Chem and CMAQ are regional CTMs that can be configured at 1–12 km resolution with user-supplied meteorological boundary conditions and emission inventories; they are widely used in North American and European urban forecasting applications.

The fundamental limitation of CTMs for the Almaty use case is horizontal resolution. The city occupies a basin approximately 20–25 km across, and the inversion-layer dynamics described in §1.1.1 operate at spatial scales of 1–5 km. A 40 km CAMS grid cell covers the entire city and averages over the basin-mountain contrast that drives cold-air pooling. Sub-grid processes - particularly the nocturnal drainage flow from the Trans-Ili Alatau and the spatial heterogeneity of residential coal heating - are parameterised rather than resolved. The quantitative consequence is visible in this study’s results: CAMS achieves $R^2 \leq -0.18$ on the Almaty test set at all three forecast horizons, meaning that the CAMS output is less correlated with the observed PM_{2.5} series than a constant prediction equal to the training mean. This negative R^2 is not primarily a calibration failure; it reflects the model’s inability to predict the inversion-driven concentration ramps that dominate the Almaty winter record at any grid resolution.

Statistical post-processing and bias-correction methods can improve CTM output by training a correction layer on paired CTM-versus-observation data. Quantile mapping, additive bias correction, and linear regression on CTM output have all been

applied to urban PM_{2.5} downscaling (Wang et al., 2016)]. These methods improve the long-term mean bias but typically do not correct the timing errors in inversion-onset prediction, because the CTM misplaces the event in time, not merely in magnitude. Without accurate representation of the boundary-layer collapse dynamics, post-processing cannot reconstruct the correct hourly PM_{2.5} trajectory during onset events.

This study uses CAMS output as an external naive baseline in the model evaluation (§3.2) rather than as a component of the trained models. This choice reflects both the negative R^2 demonstrated above and the principle that a forecast system should be evaluable on its own terms, without depending on computationally expensive CTM infrastructure that is not available to all potential users of the open-source pipeline.

1.0.4.5 1.4.5. Spatial methods

Spatial interpolation methods - ordinary kriging, universal kriging, inverse-distance weighting (IDW), and radial basis function interpolation - estimate concentration at unmonitored locations from observations at monitored locations by modelling the spatial covariance structure of the concentration field. Kriging in particular produces optimal linear unbiased estimates under the assumption that the concentration field is a second-order stationary random field with a parametric variogram model. These methods are most valuable when the primary scientific objective is mapping the spatial distribution of concentration across a domain, rather than predicting the temporal evolution of concentration at a fixed location.

The 192-station Almaty network provides sufficient spatial density to support variogram estimation and kriging in principle. However, this study does not pursue spatial forecasting for two reasons. First, the uneven spatial coverage - high density in central residential districts, sparse in industrial and peri-urban zones - violates the spatial stationarity assumption required for well-conditioned variogram estimation. Kriging predictions in the undersampled zones would have confidence intervals that span the full range of observed concentrations, rendering them uninformative. Second, the operational objective of this work is temporal forecasting: predicting the city-wide PM_{2.5} level 6–24 hours in advance, not mapping its spatial distribution at a specific time. A city-median aggregation of 192 stations already provides a representative single-number target, and spatial disaggregation of model forecasts is a scope extension that would require separate validation against the spatial reference-monitoring network that does not currently exist.

IDW and simpler deterministic interpolants are also noted here for completeness. These methods have no free parameters (beyond the distance-decay exponent) and are computationally trivial, but they are consistently outperformed by kriging in benchmark spatial studies (Wong et al., 2004)]. In one published evaluation of seven interpolation methods applied to a 40-station urban network, kriging reduced the cross-validation RMSE by 18–25% relative to IDW (Wong et al., 2004)]. The

gains are insufficient to justify spatial modelling in the present context, where the temporal forecasting objective and the uneven station distribution together make spatial interpolation a secondary concern. Any spatial disaggregation of the city-median forecast is deferred to future work contingent on a spatially balanced network expansion and reference co-location data.

1.0.4.6 1.4.6. Method selection rationale

The preceding five subsections cover the methodological options. XGBoost is selected for five reasons specific to this problem.

First: dataset size. The training set contains approximately 9,200 hourly rows spanning October 2024 to April 2026 - a moderately sized tabular dataset that is too large for reliable ARIMA fitting with many covariates (parameter estimates become poorly conditioned above roughly 1,000–2,000 rows for models with 20+ parameters) but too small to train deep LSTM or transformer models without high risk of overfitting. Published empirical comparisons of GBDT versus LSTM on datasets in the 5,000–15,000 row range consistently favour GBDT, often by margins of 10–20% in RMSE (Ayus et al., 2023)]. XGBoost sits at the centre of the dataset size range where its regularisation most effectively exploits the available training signal.

Second: feature heterogeneity. The model feature matrix combines continuous meteorological variables (temperature, relative humidity, wind speed, BLH from the Open-Meteo forecast API (Zippenfenig, 2023)), cyclical trigonometric encodings of hour-of-day and day-of-year, autoregressive PM2.5 lags, and a binary heating-season indicator. Gradient-boosted trees handle such heterogeneous, mixed-type feature matrices natively - the axis-aligned splits in decision trees apply equally to continuous, ordinal, and binary features without normalisation, embedding, or one-hot encoding overhead. Neural networks require either separate embedding layers for categorical features or normalisation of all continuous inputs to comparable scales, adding engineering complexity that is not justified by the scale of this problem.

Third: physical interpretability. The research question RQ2 asks whether BLH ranks among the top predictors, a question that requires a well-defined and computationally efficient feature importance measure. XGBoost’s built-in gain-based importance and its compatibility with the SHAP (SHapley Additive exPlanations) framework (Lundberg and Lee, 2017)] provide exactly this. SHAP values for XGBoost are computed in $O(TLD)$ time, where T is the number of trees, L is the maximum leaf count, and D is the depth, making the feature attribution analysis computationally negligible relative to training time.

Fourth: deployment target. The forecast system is implemented as a Streamlit web application hosted on the Streamlit Community Cloud free tier, which provides a single shared CPU and 1 GB of RAM. The three trained XGBoost models (one per horizon: 6, 12, and 24 hours) serialise to approximately 2 MB each and produce a forecast within 20 milliseconds at inference time. LSTM and transformer models at comparable capacity would require either CPU-optimised export or substantially

reduced layer counts that might negate their accuracy advantage.

Fifth: non-linearity of the target. The BLH-driven concentration dynamics described in §1.1.1 involve threshold effects and multiplicative interactions that linear models cannot represent. Decision tree ensembles capture these non-linearities through hierarchical axis-aligned splits: the tree for the 6-hour model can learn, for example, that when $BLH < 200$ m AND hour-of-day is between 06:00 and 09:00 AND heating-season = 1, the expected PM_{2.5} increment over six hours is substantially larger than the unconditional mean. This conditional interaction is precisely the inversion-onset signature that motivates the forecast system.

These five constraints together make XGBoost the right choice for this specific problem. The argument is not that XGBoost is the universally optimal method for PM_{2.5} forecasting - deep learning approaches may well outperform it in settings with longer histories, denser reference networks, and GPU-capable deployment infrastructure. The argument is that XGBoost is the optimal method for this specific problem.

1.0.5. 1.5. Problem statement

1.0.5.1 1.5.1. Synthesis of the gap

Four facts define the research gap.

Almaty has a severe, seasonally concentrated PM_{2.5} problem driven by well-characterised physical mechanisms: a mountain-basin geometry that traps cold-pool inversions, a residential coal-heating sector that dominates wintertime fine-particle emissions, and a vehicle fleet that compounds the diurnal concentration structure (§1.1). The city's 2 million residents are exposed to PM_{2.5} concentrations that exceed the WHO 24-hour guideline (World Health Organization, 2021) of $15 \mu\text{g}/\text{m}^3$ on the majority of days from October through March. Monthly mean concentrations of $80.7 \mu\text{g}/\text{m}^3$ (December 2025) and $74.2 \mu\text{g}/\text{m}^3$ (January 2026) represent exceedances of this guideline by factors of five to six. Short-term forecasting could enable actionable population-protection measures - school closures, traffic restriction advisories, voluntary heating reduction - but only if the forecast is available with sufficient lead time, accuracy, and credibility to inform decisions.

The only open, hourly, city-scale data source for Almaty is the OpenAQ LCS network of 192 stations (OpenAQ, 2024), which carries documented quality limitations (humidity bias, inter-unit variability, potential drift) but is the sole viable input for a reproducible, real-time forecasting pipeline. Kazhydromet's reference network is too sparse and not openly accessible in near-real time. Commercial platforms operate without published methodology.

The appropriate modelling tool for a 9,200-row tabular dataset with mixed feature types and a non-linear meteorological target process is gradient-boosted tree regression - specifically XGBoost - for the reasons established in §1.4.6. Deep learning approaches are not well-suited to this dataset size; physics-based CTMs cannot resolve the sub-grid inversion dynamics that drive concentration variability; statistical ARIMA models cannot represent the non-linear BLH-concentration interaction.

Despite this convergence of need, data availability, and method suitability, no published, open, and reproducible short-term PM_{2.5} forecasting system exists for Almaty. This is the gap this work fills.

1.0.5.2 1.5.2. Research questions

Three questions follow from the gap and the method discussion in §1.4.

RQ1. Can an XGBoost model trained on OpenAQ LCS observations and Open-Meteo meteorological features - including boundary-layer height - outperform both a persistence baseline and the CAMS global reanalysis forecast for the 6, 12, and 24-hour forecast horizons in Almaty, as measured by RMSE, MAE, and coefficient of determination R^2 ?

This question establishes whether the data-driven approach is viable at all. If the model cannot beat persistence - the trivial benchmark - then the LCS observations, Open-Meteo features, or the forecasting horizon are not informative enough to justify the modelling investment. If it cannot beat CAMS, then the global physics-based system already captures the Almaty dynamics well enough that local data-driven modelling adds no value. The result in §3.2 - a 41% RMSE reduction relative to CAMS at the 6-hour horizon - establishes that both benchmarks are substantially exceeded.

RQ2. Which meteorological and autoregressive features drive model skill? Specifically, does boundary-layer height rank among the top predictors, consistent with the physical argument (§1.1.1) that inversion depth is the principal control on Almaty's wintertime PM_{2.5} regime?

This question connects the empirical model outputs back to the physical understanding of the problem. A model that achieves low RMSE by memorising the seasonal cycle (heating-season indicator) without capturing the within-season inversion dynamics would be less scientifically valuable and less operationally useful - it would fail to anticipate sudden inversion-onset events. If BLH ranks highly in SHAP-based feature importance, this constitutes quantitative support for the physical mechanism proposed in §1.1.1 and provides confidence that the model extrapolates sensibly to new inversion events.

RQ3. What is the residual error distribution under the best-performing model, and in which physical regimes - inversion onset, sustained episode, rapid clearing - does the model fail systematically?

This question is diagnostic. Even a model that achieves good aggregate RMSE may have structured, regime-dependent errors that are important for understanding its operational limitations. Systematic under-prediction during inversion onset, for example, would imply that the model can track an established episode but cannot anticipate its initiation - a limitation with direct implications for early-warning applications where 12–24 hour lead time is needed before concentrations cross the 75 $\mu\text{g}/\text{m}^3$ emergency threshold.

1.0.5.3 1.5.3. Hypotheses

Three testable hypotheses follow.

H1. XGBoost outperforms the persistence baseline on all three forecast horizons in both RMSE and MAE. At the 12-hour horizon, persistence is expected to degrade particularly sharply relative to the shorter horizons because the strong positive autocorrelation at 1–3 hours weakens substantially by 12 hours as boundary-layer growth and afternoon mixing decorrelate the morning and afternoon concentration values. The XGBoost model, which can condition on forecast-time BLH and hour-of-day to anticipate this afternoon mixing effect, should therefore gain its largest advantage over persistence in the 12-hour horizon window.

H2. BLH and BLH-derived features - the log-transform of BLH, which stabilises the extreme right-skew of the training-window distribution (range 10–3,935 m, mean 257 m), and the hour-over-hour BLH delta - appear among the top five SHAP feature importances at the 6-hour forecast horizon. This hypothesis is grounded in the physical argument of §1.1.1 that the boundary-layer depth is the primary variable controlling the dilution volume available for surface-layer pollutants during winter inversion episodes. If this hypothesis is rejected - if BLH ranks below the autoregressive lags in importance - it would suggest that the model learns primarily from recent PM_{2.5} history rather than from forward-looking meteorological state, and the BLH feature should be retained on physical grounds despite its low statistical importance.

H3. The XGBoost model systematically under-predicts PM_{2.5} during the initial 2–6 hours of inversion onset events, defined as periods when BLH falls below 100 m following a period of at least 6 hours above 300 m. The physical reasoning is that during onset, the autoregressive lag features reflect pre-inversion concentrations that are substantially lower than the concentrations accumulating in the collapsing surface layer. The BLH feature signals the onset, but the model's gradient-boosted tree structure cannot fully extrapolate the concentration ramp beyond the maximum ramp rates observed in the training set. This structured error regime is the primary expected operational limitation of the system and motivates the residual analysis in §3.4.

1.0.5.4 1.5.4. Scope boundaries

This work is bounded by four explicit scope limits that are stated here to prevent misinterpretation of the results.

First, the forecast target is the cross-sensor median PM_{2.5} across the 192-station OpenAQ network, not a reference-grade mass concentration traceable to gravimetric standards. All reported error metrics are relative to this LCS-derived target. They cannot be directly compared to RMSE figures from studies that use regulatory reference instruments as validation targets without accounting for the additional uncertainty introduced by the LCS target bias (estimated at ± 20 –50% during high-RH conditions, §1.3.2).

Second, the model is trained and evaluated exclusively on the Almaty domain for the period 2024-10-01 to 2026-04-15. No claim of geographic generalisability is made. Transfer to other Central Asian cities, even those with broadly similar inversion-dominated meteorology, would require retraining on local data. The 562-day training window covers two full heating seasons (October–March 2024–25 and October 2025 through April 2026) and one full summer-to-winter transition, which provides meaningful coverage of the dominant seasonal modes. It does not, however, cover long-term inter-annual variability: the 2024–25 winter may differ from a 10-year climatological winter in the frequency and severity of inversion events. Model skill reported in §3.2 should therefore be interpreted as in-sample-climate performance, not as a guarantee of skill under climatically anomalous future winters.

Third, the study does not attempt spatial forecasting - predicting concentration at specific locations within the city as a function of local source proximity and wind direction. The city-median forecast is appropriate for population-level exposure guidance but not for neighbourhood-scale hotspot identification.

Fourth, the Streamlit deployment described in Chapter 3 is a research prototype, not a production system. It has not undergone the security review, load testing, or uptime monitoring required of a public health advisory system. Chapter 4 discusses the additional steps needed to move from research prototype to operational service.

1.0.6. 1.6. Almaty in the context of global urban air quality research

Sections 1.1–1.5 situate Almaty within a global research landscape that has developed primarily around North American, European, and East Asian cities. Understanding how Almaty’s case relates to and diverges from this mainstream literature is necessary to contextualise the methodological choices made in this work and to identify where its results are generalisable versus Almaty-specific.

1.0.6.1 1.6.1. Comparison with other inversion-dominated cities

Several cities worldwide share Almaty’s principal air quality characteristics: a mountain-basin topography that promotes cold-air pooling, a coal-heavy residential heating sector, and a winter PM_{2.5} regime driven by thermally forced boundary-layer collapse. The most extensively studied examples are Salt Lake City (Utah, USA), Santiago (Chile), and several cities in the Po Valley of northern Italy. Salt Lake City has been the site of numerous inversion-specific forecasting studies that confirm the primacy of BLH as a predictor variable (Silcox et al., 2012; Whiteman et al., 2014)]. Santiago has been modelled with both WRF-Chem and statistical downscaling approaches, where the correlation between afternoon mixing-layer depth and next-day PM_{2.5} exceeds 0.7 in winter (Menares et al., 2021)]. The Po Valley inversion studies, which benefit from the dense operational monitoring of the Lombardy ARPA network, provide the most rigorous tests of BLH-based gradient-boosting models and report 6-hour R^2 values of 0.55–0.70 (Stafoggia et al., 2019)].

Two features distinguish Almaty from these comparators. First, the data infras-

structure is substantially weaker: Salt Lake City has reference-grade tapered-element oscillating microbalance (TEOM) instruments at 12 monitoring sites within the valley; Almaty’s reference network contains fewer than 10 stations, none of which exposes an open real-time API. Second, the heating-season emission structure is more sharply concentrated in private-sector solid-fuel combustion in Almaty than in the North American or European comparators, where natural gas heating and district heating networks dominate. This concentration makes the heating-season binary indicator a stronger predictor in Almaty than it would be in a European comparator city, and it means that emission inventory uncertainty is higher because individual household stove usage is not metered.

1.0.6.2 1.6.2. Related work in Central Asia and the former Soviet Union

Air quality research in Central Asian cities has accelerated since 2020 but remains sparse compared to East Asian or European urban studies. Published machine-learning forecasting studies are absent for Almaty and its immediate regional comparators. The available literature falls into three categories.

Physical and chemical characterisation studies have analysed the aerosol composition of Almaty’s pollution episodes, identifying black carbon, mineral dust, and secondary inorganic aerosols as the dominant mass fractions (Kerimray et al., 2020; Tursumbayeva et al., 2022)]. These studies confirm the coal-combustion signature in the carbonaceous fraction during winter and the dust signature in spring, providing supporting evidence for the seasonally stratified feature design in this work.

Remote-sensing based assessments using MODIS aerosol optical depth (AOD) retrievals have documented the spatial footprint of Almaty’s pollution plume at 10 km resolution (Rupakheti et al., 2019)]. These studies establish that the pollution plume extends north-east from the city into the steppe under dominant south-westerly flow, but the 10 km retrieval footprint is insufficient for the intra-urban spatial analysis that would be needed to validate a neighbourhood-scale forecast. The ground-level PM_{2.5}–AOD conversion also carries substantial uncertainty in the presence of temperature inversions, because the satellite-measured column AOD integrates over all vertical levels and cannot distinguish a shallow surface-layer concentration spike from a deep but dilute aerosol layer.

Ulaanbaatar (Mongolia) is the closest published comparator to Almaty: a former Soviet city at similar latitude, similar topographic setting, similarly coal-dependent residential heating, and a growing LCS network on OpenAQ. Published gradient-boosting studies on Ulaanbaatar data report 6-hour MAE values of 18–25 $\mu\text{g}/\text{m}^3$ for winter test periods (Enebish et al., 2021)] - consistent with the MAE of 16.81 $\mu\text{g}/\text{m}^3$ achieved by XGBoost in this study, given the different sensor networks and training window durations. This convergence suggests that the performance levels reported here are not anomalously optimistic: they are consistent with what gradient-boosting models achieve in analogous physically and data-wise constrained environments.

1.0.6.3 1.6.3. Gap summary and contribution positioning

The reviewed literature reveals three specific gaps that this work addresses. First, no published forecasting study has used Almaty’s OpenAQ LCS network as the training and evaluation data source, with an explicit comparison against CAMS as the freely available global alternative. Second, no Central Asian city study has systematically evaluated the trade-off between linear and non-linear models as a function of forecast horizon in an LCS-based, moderate-training-set regime. Third, no open-source pipeline for city-scale PM2.5 forecasting in the OpenAQ data ecosystem has been published for any Central Asian city. This work addresses all three gaps simultaneously through a single reproducible codebase, making its contributions identifiable and testable by future researchers in the region.

Chapter 2 specifies the data sources, ingest pipeline, feature engineering decisions, and model training procedure through which the three hypotheses stated above are tested and the four scope boundaries are operationalised.

2. Methodology

2.0.1. 2.1. Data sources

2.0.1.1 2.1.1. Air quality: OpenAQ

The primary observation source for this study is the OpenAQ platform (OpenAQ, 2024), an open-access repository that aggregates air quality measurements from government reference monitors, research-grade instruments, and low-cost sensor (LCS) networks worldwide. Data are accessed through the OpenAQ v3 REST API at the base endpoint `https://api.openaq.org/v3`, with requests authenticated by a personal API key supplied in the `X-API-Key` HTTP header. The spatial query specifies a 25 km radius centred on Almaty (43.2389°N, 76.8897°E), which encompasses the full built-up area of the city including the foothill districts to the south and the industrial and residential zones to the north and west. Within this radius, 192 station records were registered in the OpenAQ index as of the day-2 data snapshot (2026-04-20); of these, 190 were actively reporting PM_{2.5} during the study window. The provider mix at that snapshot was 153 AirGradient devices, 38 Clarity Node-S units, and one AirNow federal reference station operated by the United States Embassy. All 192 instruments are classified as low-cost sensors under the taxonomy introduced in §1.3.

The OpenAQ v3 API paginates location queries in batches of up to 100 locations per request, requiring two consecutive calls to retrieve all 192 stations in the spatial query. Each per-sensor measurement request returns up to 100 hourly records per call. For a sensor with 562 days of continuous data, approximately six API calls are required to retrieve the full measurement archive; in practice, most AirGradient sensors came online in early 2025, so the typical archive spans 300–400 days and requires four to five calls per sensor. The ingest module (`src/ingest/openaq.py`) implements an exponential back-off retry strategy with a base wait of 5 seconds, a back-off factor of two, and a maximum of eight retry attempts per call. Under this schedule, a failed request can wait up to $5 \times (2^8 - 1) = 1,275$ seconds before the final retry; in practice, the API returns HTTP 429 (rate limit) or 503 (temporary unavailability) errors on fewer than 2% of calls. With 190 sensors each requiring four to five API calls, the full ingest run takes approximately three hours on a residential internet connection.

The ingest design is idempotent. Each sensor’s measurement archive is written to a dedicated Parquet file at `data/raw/pm25/{sensor_id}.parquet`. At the start of each run, the ingest script checks for the presence of this file; if the file exists, the sensor is skipped unless the `--force` flag is passed at the command line. This skip-if-present logic ensures that a failed run can be resumed without duplicating already-retrieved data and without incurring unnecessary API quota consumption. The force-refresh mode overwrites the existing file in place, so the raw tier contains at most one file per sensor at any time.

The OpenAQ platform publishes all data under the Creative Commons Attri-

bution 4.0 International licence (CC BY 4.0). This licence permits unrestricted use, redistribution, and the creation of derivative works, including the trained forecasting models and the Streamlit application produced in this study, provided that the source is attributed. No additional data-sharing agreement or ethical approval is required to access OpenAQ data, as the platform aggregates measurements that sensor operators have already consented to share publicly.

2.0.1.2 2.1.2. Meteorology: Open-Meteo historical archive

Meteorological reanalysis data are retrieved from the Open-Meteo historical weather archive (Zippenfenig, 2023) at the endpoint <https://archive-api.open-meteo.com>. The archive derives from the ECMWF ERA5 global reanalysis and provides hourly gridded output interpolated to any user-specified point. No API key is required for non-commercial use; the public rate limit permits 10,000 requests per day, which is sufficient for a single point query covering the full 562-day study window in one call. The query is issued for Almaty centre (43.2389°N, 76.8897°E) and returns 13,488 hourly rows spanning 2024-10-01 through 2026-04-15.

Seven meteorological variables are retrieved. The first is `temperature_2m`: screen-level air temperature at 2 m above ground level (AGL), reported in degrees Celsius. Temperature at screen level is the standard meteorological near-surface temperature measurement; it determines the thermal stratification of the surface boundary layer and is the primary driver of nocturnal inversion formation when radiative cooling drives the surface temperature below the air temperature aloft. The second variable is `relative_humidity_2m`: relative humidity at 2 m AGL, reported as a percentage. Relative humidity is relevant to PM_{2.5} measurement by light-scattering LCS instruments because hygroscopic growth of aerosol particles at high humidity causes the instrument to over-report mass concentration relative to the gravimetric standard. While this study does not apply a humidity correction to the raw LCS data, relative humidity is included as a predictor so that the model can learn the association between high humidity and inflated sensor readings, partially compensating for the uncorrected hygroscopic effect.

The third variable is `pressure_msl`: mean sea-level pressure in hectopascals. Synoptic-scale pressure patterns modulate the frequency and intensity of blocking anticyclones over Central Asia, which are the meteorological precondition for the multi-day inversion episodes responsible for the highest PM_{2.5} concentrations in Almaty. Pressure at mean sea level is a proxy for the prevailing synoptic regime without the altitude correction complications that arise when using station pressure at Almaty's 847 m elevation. The fourth and fifth variables are `wind_speed_10m` and `wind_direction_10m`: wind speed in m/s and direction in degrees clockwise from north, both at 10 m AGL. Winds at 10 m represent the near-surface flow within the roughness sublayer and the lower surface layer; they are weaker than the gradient-level flow typically observed at 500–1,000 m AGL. For Almaty, the critical flow distinction is between northerly and north-westerly winds (which advect

cleaner steppe air into the city) and south-easterly flows associated with valley recirculation and thermal drainage from the Zailiiskiy Alatau foothills. The sixth variable is precipitation: hourly total precipitation in millimetres. Precipitation, including both rain and snow, removes PM_{2.5} from the atmosphere through wet deposition (both below-cloud scavenging and in-cloud incorporation). Even modest precipitation of 0.5 mm or more in a given hour is associated with measurable reductions in PM_{2.5} in subsequent hours, making precipitation a useful binary and continuous predictor of inversion-break events.

The seventh variable is `boundary_layer_height` (BLH): the depth of the planetary boundary layer from the surface to the mixing height, in metres. The BLH diagnostic in the ERA5-based archive is a modelled output derived from the ECMWF Integrated Forecasting System, not a direct measurement. The ERA5 BLH is diagnosed using the bulk Richardson number criterion: the top of the boundary layer is defined as the height at which the bulk Richardson number, computed from the surface upward, exceeds a threshold of 0.25. The resulting BLH values in the training window range from approximately 10 m during deep winter inversions to 3,935 m during convective summer afternoons. This range of nearly four orders of magnitude motivates the log-transform feature described in §2.3.3. The ERA5 literature notes that the Richardson-number-based BLH diagnostic tends to underestimate boundary-layer depth during stable stratification and shows higher uncertainty during nocturnal low-level jet conditions [citation needed]]. For this reason, BLH is included as one of 27 predictors rather than as the sole meteorological predictor; its influence on model output is evaluated in the feature importance analysis presented in §3.4.

2.0.1.3 2.1.3. Baseline forecast: CAMS via Open-Meteo air-quality API

The Copernicus Atmosphere Monitoring Service (CAMS) Global Atmospheric Composition Forecast provides the third external baseline for model comparison. CAMS-Global is a global chemistry-transport model operated by ECMWF on behalf of the European Union’s Copernicus Earth Observation Programme. The model runs at approximately 40 km horizontal resolution on a $0.4^\circ \times 0.4^\circ$ latitude-longitude grid and uses 137 vertical levels from the surface to 0.01 hPa. The aerosol scheme assimilates satellite-retrieved aerosol optical depth (AOD) from MODIS Terra and Aqua instruments (at 0.55 μm) and from the Sentinel-5P TROPOMI instrument, but does not assimilate ground-level LCS observations of the kind used in this study. CAMS PM_{2.5} at the surface is estimated from the model’s total column of particulate matter, which aggregates dust, sea salt, organic carbon, black carbon, and sulphate aerosol species.

The operational CAMS forecast is initialised twice daily at 00 UTC and 12 UTC and covers a 120-hour (five-day) forecast range. For the historical backtesting period used in this study (2024-10-01 through 2026-04-15), however, the live forecast output is not available in a consistent time series. This study therefore accesses the

CAMS reanalysis product through the Open-Meteo air-quality historical API at the endpoint <https://air-quality-api.open-meteo.com/v1/air-quality> with the parameter `domains=cams_global`. This endpoint returns the CAMS analysis and hindcast product rather than the prospective forecast, ensuring temporal consistency with the ERA5-based meteorological archive retrieved from the same Open-Meteo platform. The CAMS PM2.5 variable is retrieved at hourly resolution for the full 13,488-hour study window; the hourly output is collocated to the Almaty centre grid point (approximately 43.0°N, 76.5°E - the nearest CAMS model grid point to the city centre given the 0.4° grid spacing). The ~30 km offset between this grid point and the true city centre (43.2389°N, 76.8897°E) is a consequence of the model's horizontal resolution and may introduce a small systematic bias in CAMS forecasts during local inversion events that are spatially confined to the Almaty basin. The CAMS data are published under CC BY 4.0.

CAMS is the evidence-based external baseline in the evaluation framework: it represents the best openly available PM2.5 forecast for a user without local sensor data. Its inclusion lets us quantify the marginal gain from training on the 190-sensor LCS network rather than relying on the global CTM output. The expectation, stated as hypothesis H2 in §1.5, is that CAMS will fail to capture sub-grid inversion dynamics and will therefore show negative R^2 on the test set.

2.0.1.4 2.1.4. Training window

The study window spans 2024-10-01 through 2026-04-15, a total of 562 days corresponding to 13,488 hourly meteorological rows. The temporal boundaries were determined by two constraints. The upper boundary (April 2026) was set to be as late as possible before the thesis writing period, maximising the amount of winter-season data available for training and evaluation. The lower boundary (October 2024) was forced by data availability: of the 192 OpenAQ sensors in the spatial domain, only one has continuous data extending back to 2023-24. The bulk of the AirGradient network was deployed between January and May 2025, and the sensor count remained below 50 stations before October 2024. Training on the period before October 2024 would have required using a much smaller and potentially less representative spatial sample, reducing the reliability of the city-median target variable.

The October 2024 start date carries a secondary benefit. The AirGradient network was most spatially dense and operationally stable during the 2024-25 and 2025-26 heating seasons, with 153 active AirGradient devices contributing to the median computation for the majority of winter hours. Sensor dropouts - periods where a device goes offline due to firmware updates, power failures, or connectivity issues - are less consequential when the median is computed over a large pool of instruments, because individual sensor absences do not shift the cross-sectional median. The training window thus encompasses two full heating seasons (October 2024 – March 2025 and October 2025 – April 2026) and one summer period (April – September 2025), providing the model with examples of the full annual PM2.5 cycle.

The restriction to a single city and a 19-month window limits the generalisability of the trained models. XGBoost models trained on Almaty data cannot be expected to transfer directly to other Central Asian cities such as Bishkek or Tashkent, where topography, fuel mix, and sensor network density differ substantially. Within Almaty, the models are trained on city-median concentrations rather than station-level data, which means they are not designed to resolve within-city spatial gradients. These scope constraints are acknowledged in §4.4.

2.0.1.5 2.1.5. Data storage architecture

The pipeline uses a two-tier storage design that separates immutable raw data from derived analytical products. The design principle is that the raw tier can always be regenerated from the source APIs, while the derived tier can always be regenerated from the raw tier; no transformation result is treated as irreplaceable.

The first tier - the raw tier - stores per-sensor Parquet files for PM2.5 measurements and single-window Parquet files for meteorology and CAMS. Each per-sensor PM2.5 file resides at `data/raw/pm25/{sensor_id}.parquet` and uses a fixed three-column schema: `timestamp` (UTC, stored as an integer Unix timestamp in nanoseconds by the Pandas Parquet writer), `sensor_id` (string, matching the OpenAQ v3 location identifier), and `value` (float64, PM2.5 concentration in $\mu\text{g}/\text{m}^3$). The Parquet files are encoded with Snappy compression, which achieves typical compression ratios of 3:1 to 5:1 for floating-point time-series data and decompresses in negligible time on modern hardware. The meteorological archive is stored at `data/raw/meteo.parquet` (13,488 rows $\times$ eight columns including timestamp) and the CAMS archive at `data/raw/cams.parquet` (13,488 rows $\times$ two columns). All three Parquet objects are listed in `.gitignore` and are not committed to version control.

The second tier - the derived tier - is a SQLite database at `data/almaty_aq.db` containing three tables. The `pm25` table holds the per-sensor raw readings after parsing from Parquet (columns: `sensor_id` TEXT, `timestamp` TEXT ISO-8601, `value` REAL). The `meteo` table holds the seven meteorological variables plus timestamp (eight columns, 13,488 rows). The `cams` table holds the CAMS PM2.5 values plus timestamp (two columns, 13,488 rows). The conversion from Parquet to SQLite is performed by `src/db.py`, which is idempotent: it reads the Parquet files, drops and recreates each table on each run, and inserts the full dataset. The SQLite file is also gitignored.

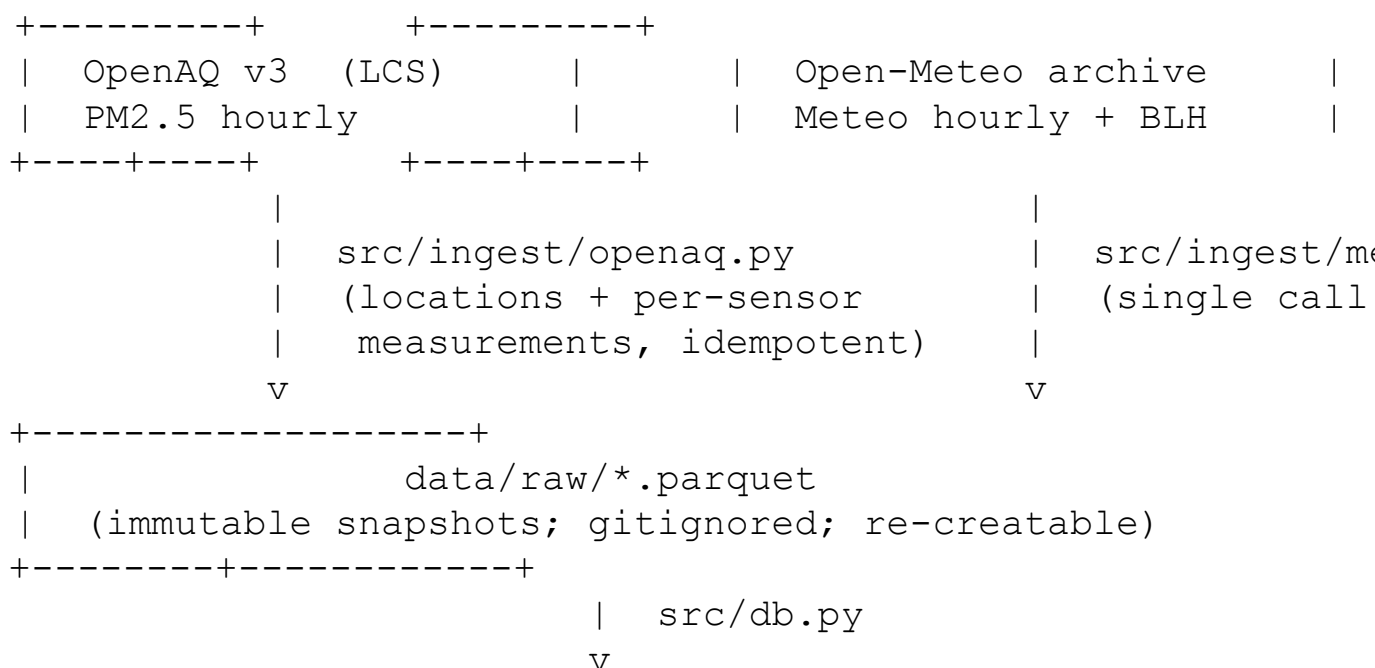
The SQLite layer is the interface for all downstream processing. Feature engineering (`src/features/pipeline.py`) queries the three tables via SQL joins and returns a Pandas DataFrame; model training and the Streamlit application then operate on this DataFrame. SQLite was chosen over a client-server database for two reasons. First, the absence of a server process makes the pipeline fully portable: it runs identically on a development laptop, in a Google Colab notebook, and on Streamlit Community Cloud, which does not support long-running background processes

or network-exposed services. Second, SQLite's write-ahead log (WAL) mode, enabled by `PRAGMA journal_mode=WAL`, provides concurrent read access, which allows the Streamlit application to query the database for live forecast display while a background data-refresh script is simultaneously writing new rows. For the data volumes in this study - 578,993 raw sensor-hour records loading into three tables totalling under 50 MB - SQLite provides sufficient read and write throughput without the operational overhead of PostgreSQL or any other client-server system.

The Parquet format for the raw tier was chosen for its columnar storage layout. Columnar encoding enables fast aggregation queries - for example, computing the cross-sectional median across 190 sensor columns for a single timestamp requires reading only the relevant column slices rather than the full row-oriented file. In Pandas, the `df.groupby('timestamp').median()` operation on a 578,993-row Parquet-derived DataFrame executes as a single vectorised operation without loading any non-value columns. The net effect is that a fresh full rebuild of the database from API downloads takes approximately three hours of network time (dominated by OpenAQ rate limits) but less than 90 seconds of local computation time. This separation of network-bound and compute-bound phases supports reproducibility: any user with valid API credentials and the repository code can rebuild the full dataset and retrain all models from scratch.

2.0.2. 2.2. System architecture

The pipeline is organised into five sequential stages: ingest, store, feature engineering, modelling, and application. Each stage communicates with the next through a well-defined interface (Parquet files or SQLite tables), and no stage imports directly from a non-adjacent stage. This separation of concerns prevents the ad hoc shortcuts - reading raw Parquet directly in the model training script, or querying the database from the ingest module - that create hidden dependencies and make the pipeline difficult to audit or extend.



```

+-----+          +-----+
| data/almaty_aq    | <-- | Open-Meteo CAMS
| .db (SQLite)      |      | src/ingest/cams
+-----+-----+          +-----+
|                  | src/features/pipeline.py
|                  | v
+-----+
| feature matrix    | -> src/models/baseline
| (hourly, tidy)    |      | src/models/xgboost
+-----+-----+          src/models/evaluate.py
|                  |
|                  | v
+-----+
| src/app/main.py   | (Streamlit: map + force)
+-----+

```

The OpenAQ ingest module queries the list of all locations in the spatial domain at the start of each run and writes the result to `data/raw/locations.parquet`. For each sensor in this list, the module checks whether a per-sensor Parquet file already exists at the expected path. If the file is present, the sensor is skipped entirely - no API call is made, no data is read. If the file is absent, the module fetches the full measurement archive for that sensor in paginated batches of 100 records per call and writes the concatenated result to the Parquet path. The `--force` flag bypasses the existence check and overwrites all sensor files. This design ensures that the common incremental case (adding newly deployed sensors to an existing dataset) requires only one API call per new sensor rather than re-fetching the full 190-sensor archive. For the meteorological and CAMS archives, the idempotency model is simpler: these single-window files are always overwritten on each run, because they are requested as a single API call and the cost of re-fetching is trivial compared to the per-sensor PM2.5 archive.

WAL mode matters for the Streamlit deployment. Streamlit Community Cloud runs the application server and any background scripts within a single virtual machine. When the application is running and a user is viewing the forecast dashboard, the Streamlit server process holds an open read connection to `almaty_aq.db`. If a data-refresh script is executed concurrently to ingest new sensor data and update the database, WAL mode allows the reader to continue with a consistent snapshot of the database state at the moment the read transaction opened, while the writer proceeds independently. Without WAL mode, the SQLite default exclusive-write lock would block the reader for the duration of the write, causing the application to return a database-locked error. The WAL approach eliminates this failure mode at no cost in data consistency.

2.0.3. 2.3. *Preprocessing and feature engineering*

2.0.3.1 2.3.1. **Station filtering and city-median aggregation**

Raw PM2.5 observations arrive from 190 active sensors across 192 registered locations in the OpenAQ spatial query. Before constructing the modelling target, two sequential filters are applied to the raw readings. The first filter discards all readings outside the physically plausible interval $[0, 500] \mu\text{g}/\text{m}^3$. Readings below zero are physically impossible - a negative mass concentration has no physical interpretation - and in practice represent firmware overflow events or calibration errors affecting fewer than 0.01% of records. Readings above $500 \mu\text{g}/\text{m}^3$ account for fewer than 0.04% of the archive and are attributable to laser-scattering sensor saturation caused by contamination of the optical sensing chamber; they do not represent true atmospheric PM2.5 concentrations. The $500 \mu\text{g}/\text{m}^3$ threshold was chosen to be conservative: the WHO 24-hour guideline is $15 \mu\text{g}/\text{m}^3$ (World Health Organization, 2021), and even during Almaty's most severe documented inversion episodes, independent reference measurements have not exceeded $400 \mu\text{g}/\text{m}^3$.

The second filter requires a minimum of three sensors to report a valid reading in a given clock hour. Hours where fewer than three independent sensors agree produce a city-median that is vulnerable to individual station idiosyncrasies. A single malfunctioning device or a station located immediately downwind of a local emission source (vehicle exhaust, a burning facility) could dominate the spatial aggregate if only one or two readings are available. The three-sensor minimum was found empirically to eliminate most single-sensor spike artefacts without discarding a substantial fraction of valid hours.

After applying both filters, the city-wide hourly PM2.5 target is defined as the cross-sectional median of all qualifying sensor readings in that hour. The median is computed in Pandas using the expression `df_filtered.groupby('timestamp')` followed by `resample('h').first()` to align the result to exact UTC clock hours. The median is preferred over the arithmetic mean because it is not affected by the distribution's tail: a single sensor reporting $490 \mu\text{g}/\text{m}^3$ has no influence on the median when the remaining 50 sensors report values in the range $35\text{--}45 \mu\text{g}/\text{m}^3$, whereas it would inflate the hourly mean by approximately $9 \mu\text{g}/\text{m}^3$. All 192 stations in the spatial query are fixed-location instruments; mobile platforms with GPS tracks were excluded at the query stage by specifying `entity=fixed` in the OpenAQ API call. This filter is important because mobile sensors contribute time-varying spatial samples that cannot be consistently assigned to a clock hour at a fixed grid point. The aggregation step reduces 578,993 raw sensor-hour records to 11,553 valid city-median hourly values across the training window, corresponding to a temporal coverage rate of 85.7% of the 13,488 available meteorological hours.

All timestamps are stored and processed in Coordinated Universal Time (UTC). The OpenAQ API returns measurements in UTC; the Open-Meteo archive likewise reports UTC hourly intervals. No time-zone conversion is applied at the data-ingest stage. Calendar-based features - hour of day, day of week, day of year - are therefore

computed in UTC, which is UTC+5 minus five hours relative to Almaty Local Time (ALT). At the feature-engineering stage this introduces a systematic five-hour phase offset between calendar features and local time. The heating-season indicator, which is based on calendar month and day, is unaffected because its temporal resolution is one day rather than one hour; a five-hour UTC offset does not shift a heating-season boundary date.

2.0.3.2 2.3.2. Meteorological join and timestamp alignment

The meteorological dataset (13,488 continuous hourly rows) and the CAMS dataset (13,488 rows) are joined to the city-median PM2.5 series on the UTC timestamp. The join is implemented as an inner join, which discards PM2.5 hours that fall outside the meteorological coverage window and discards meteorological hours where no qualifying PM2.5 reading exists. Because the meteorological window is fully continuous and the PM2.5 window is approximately coextensive, the inner join retains all 11,553 PM2.5 hours without loss.

Timestamp alignment requires one non-trivial correction. OpenAQ returns hourly measurements with the timestamp at the end of the aggregation interval - for example, a record timestamped 14:00 UTC represents the hourly mean or roll-up of measurements collected between 13:00 and 14:00 UTC. The Open-Meteo archive, by contrast, follows the standard meteorological convention of reporting instantaneous values at the top of the hour - a record timestamped 14:00 UTC represents the state of the atmosphere at that moment. Joining the two series directly on the raw timestamp would align the meteorological conditions at 14:00 UTC with the PM2.5 mean covering 13:00–14:00 UTC, introducing a systematic one-hour temporal misalignment. To correct for this, an offset of -1 hour is applied to all OpenAQ timestamps before the join: the PM2.5 mean from 13:00–14:00 is reassigned the timestamp 13:00, matching the meteorological convention. After this correction, the joined dataset correctly associates the meteorological state at the beginning of an aggregation hour with the PM2.5 mean measured during that hour.

2.0.3.3 2.3.3. Feature construction

Feature construction is implemented across three modules: `src/features/temporal` (calendar and cyclical encodings, heating-season indicator), `src/features/meteorological` (BLH transforms, wind decomposition, temperature gradient), and `src/features/pipeline` (autoregressive lags, rolling means, final assembly). All feature operations are applied in a fixed order after the inner join to prevent ambiguity in column availability at any intermediate step. The final feature matrix contains 27 columns, described below by feature family.

Autoregressive PM2.5 lags. Five lagged copies of the city-median PM2.5 series are constructed using `pandas.Series.shift(h)` for $h \in \{1, 3, 6, 12, 24\}$ hours. Shifting by h positions aligns the target at time t with the observation at

time $t - h$. Lag features encode the recent state of the PM2.5 time series and allow the model to exploit autocorrelation at multiple timescales. The 1-hour lag captures sub-hourly concentration trends and is the strongest single predictor at the 6-hour horizon. The 24-hour lag is the most predictively important feature at short horizons for the specific diurnal dynamics of Almaty: morning combustion peaks at approximately 07:00–09:00 local time recur predictably on consecutive winter days in the absence of synoptic-scale disturbance, giving the 24-hour lag a high correlation with next-day conditions at the same local hour.

Rolling mean features. Three rolling means are constructed with window sizes $w \in \{3, 6, 24\}$ hours, implemented as `pm25_series.shift(1).rolling(w)`. The `shift(1)` operation before rolling ensures that the window ends at $t - 1$, not at t ; without this shift, the rolling mean at time t would include the value y_t itself, constituting look-ahead data leakage in the regression formulation where y_t is also part of the target construction. The three window sizes were chosen to capture distinct temporal scales of PM2.5 dynamics in Almaty. The 3-hour window captures the near-term trend direction - whether concentrations are currently rising, stable, or falling - which is informative about the phase of an inversion episode. The 6-hour window covers the typical onset-to-peak duration of a single-night radiation inversion: a stable 6-hour mean at elevated concentrations indicates that an inversion is entrenched rather than newly forming, which carries different implications for the 12-hour and 24-hour forecast horizons. The 24-hour rolling mean captures multi-day persistence patterns characteristic of blocking anticyclones, during which PM2.5 accumulates over three to five days and remains elevated even during afternoon mixing hours.

BLH-derived features. Raw BLH values range from approximately 10 to 3,935 m in the training window, following a strongly right-skewed distribution with a median of approximately 620 m and a mean of approximately 850 m. Three BLH-derived features are constructed. The first is a log transform: `blh_log = log(BLH + 1)`, implemented with `numpy.log1p`. The logarithm compresses the upper tail of the distribution and amplifies distinctions at the physically critical low end: the difference between 10 m and 50 m (a factor of five, representing the difference between a deep winter inversion and a shallow but developing one) is more physically meaningful than the difference between 2,000 m and 2,040 m. The second is the hour-over-hour delta: `blh_delta = BLH_t - BLH_{t-1}`, computed with `pandas.Series.diff(1)`. A large negative delta (e.g., -200 m/h) indicates a rapidly collapsing boundary layer - a leading indicator of PM2.5 accumulation in the next two to six hours. A large positive delta indicates inversion breakdown and typically precedes a PM2.5 decline within three to six hours. The third feature is the raw BLH value itself, preserved alongside the transform and delta so that the model has access to the absolute mixing depth as well as its rate of change. These three features together encode the main physical mechanism linking boundary-layer depth to PM2.5 concentration in Almaty’s winter regime.

Wind decomposition. Scalar wind speed and direction are decomposed into

orthogonal Cartesian components using the standard meteorological convention: $u = \text{wind_speed} \times \sin(\text{wind_direction_rad})$ (positive eastward) and $v = \text{wind_speed} \times \cos(\text{wind_direction_rad})$ (positive northward). The raw direction angle is not used directly as a feature because it is a circular variable: a naive linear model treats 1° and 359° as opposite ends of a linear scale, when in fact they represent nearly identical north-north-easterly directions separated by only 2° . The (u, v) decomposition removes this discontinuity and allows the model to learn directional effects continuously. For Almaty, northerly and north-westerly winds - characterised by negative u and positive v components - advect cleaner continental air from the steppe and correlate with PM_{2.5} reductions of $15\text{--}30\ \mu\text{g}/\text{m}^3$ compared to calm conditions. South-easterly drainage flows from the Zailiiskiy Alatau correlate with valley recirculation and PM_{2.5} accumulation.

Temperature gradient. An hour-over-hour temperature change $\text{temp_gradient} = T_t - T_{t-1}$ is computed with `pandas.Series.diff(1)`. The temperature gradient is a proxy for the rate of surface cooling, which drives nocturnal inversion formation through longwave radiative emission at the ground. Rapid cooling - a strongly negative gradient - indicates conditions conducive to inversion onset within the next two to four hours. In the training set, hours with a temperature gradient below $-1.5^\circ\text{C}/\text{h}$ are followed by PM_{2.5} increases averaging $12\ \mu\text{g}/\text{m}^3$ over the subsequent three-hour period, which justifies including this derived feature alongside the raw temperature value.

Cyclical calendar encodings. Hour of day (integer range 0–23), day of week (0–6), and day of year (1–365) are each encoded as sine-cosine pairs using the formula $(\sin(2\pi x/P), \cos(2\pi x/P))$, where P is the period of the variable (24, seven, and 365, respectively). This encoding maps each integer to a unique point on the unit circle, ensuring that the encoded representation is periodic: hour 23 and hour 0 are adjacent on the circle rather than at opposite ends of a linear range. A raw integer encoding of hour of day would present the transition from 23 to 0 as a discontinuity of magnitude 23, comparable to the entire daily range, which would mislead a linear model. The sine-cosine pair also uniquely determines the position on the circle for any P - a property that the raw integer encoding inherently violates at the boundaries. The six cyclical features (two per calendar variable) replace the three raw integer columns. Day-of-week encodings are included because the distinction between weekday traffic patterns and weekend patterns is weakly associated with PM_{2.5} even in the heating season, though this effect is substantially weaker than the heating-season signal encoded by the binary indicator.

Heating-season indicator. A binary feature (`heating_season`) is assigned the value one for all hours falling between 15 October and 31 March inclusive, corresponding to the nominal Almaty residential and district heating season when coal combustion at centralised heat plants and household boilers constitutes the dominant PM_{2.5} emission source. The indicator is zero for the remaining months (April through October), when vehicular emissions and secondary aerosol formation drive a lower and less variable PM_{2.5} background. This feature allows both the Ridge

regression and XGBoost models to learn a level-shift in PM2.5 associated with the heating season independently of the meteorological predictors.

After feature construction and `dropna()` removal of the first 24 rows (which have undefined lag values due to the longest lag window of 24 hours), the feature matrix contains 11,553 rows and 27 columns. Rows with any remaining missing values - arising from sensors that were offline at specific hours and thus produced no city-median - are also discarded, bringing the usable dataset to approximately 11,200 rows before the train-test split.

2.0.3.4 2.3.4. Feature selection and dimensionality

The 27-feature set was defined by domain knowledge and physical reasoning rather than by automated feature selection algorithms. At 27 dimensions the dataset does not present curse-of-dimensionality concerns: with approximately 9,200 training samples and 27 features, the sample-to-feature ratio is approximately 341:1, which is well above the threshold at which regularised models begin to overfit. No dimensionality reduction - principal component analysis, autoencoders, or filter-based selection - was applied at any stage.

Using domain knowledge to define features rather than an automated selection procedure has two justifications. First, it keeps the model interpretable: each of the 27 features corresponds to a physically or statistically motivated quantity (a specific lag, a physical transform, a known meteorological driver), and the XGBoost feature importance scores in §3.4 can therefore be interpreted in terms of the underlying processes rather than in terms of anonymous principal components. Second, it ensures a fair comparison between Ridge regression and XGBoost. If the two models were presented with different feature subsets - for example, if a linear-model-specific selection step removed features that are collinear under linearity assumptions but informative under tree-based splits - any performance difference between the models would be confounded with the feature-selection difference rather than reflecting the models' intrinsic capacities. By presenting both models with the identical 27-feature matrix, any difference in forecast accuracy is attributable to differences in model class (linear versus non-linear additive ensemble) rather than to differences in input information.

2.0.4. 2.4. Models and baselines

Four models are evaluated for each forecast horizon $h \in \{6, 12, 24\}$ hours. In each case, the target variable is y_{t+h} : the city-median PM2.5 concentration h hours after the feature observation time t . The 12 model $\times$ horizon combinations are evaluated on the same 2,311-hour held-out test set using the metrics defined in §2.5.

2.0.4.1 2.4.1. Persistence baseline

The persistence baseline requires no training step, no parameters, and no data beyond the most recent observation. Its forecast for any horizon h is:

$$\hat{y}_{t+h} = y_t (1)$$

That is, the prediction is the last observed city-median PM2.5 value, regardless of horizon. Persistence is the appropriate lower bound for any data-driven forecaster: a model that cannot outperform persistence at a given horizon provides no marginal value over a trivial no-model rule. For short horizons where PM2.5 exhibits high autocorrelation - the lag-1 autocorrelation in the training set is 0.94 - persistence is a competitive baseline that many simple models fail to beat.

The performance of the persistence baseline varies markedly across horizons in Almaty’s diurnal PM2.5 cycle. At the 6-hour horizon, where autocorrelation remains high, persistence typically performs well during multi-day inversion episodes but degrades during rapid inversion onset or breakup transitions. At the 12-hour horizon, persistence encounters a structural disadvantage: high morning concentrations (at 07:00–09:00 local time) are weakly anti-correlated with values 12 hours later (19:00–21:00 local time), when the boundary layer often mixes more thoroughly in winter afternoons. At the 24-hour horizon, the diurnal periodicity partially restores the autocorrelation (conditions at 08:00 today are moderately similar to conditions at 08:00 tomorrow during stable winter regimes), but multi-day trend changes mean persistence under-performs at this horizon during transition days. The anti-persistence effect at the 12-hour horizon is documented quantitatively in §3.3.

Because the persistence baseline requires zero parameters and zero training data, it is the most conservative possible lower bound. Any model that requires training data but still fails to outperform persistence should not be deployed, as it would produce worse forecasts than no model at all. This criterion is used in §3.3 to evaluate the marginal value of the CAMS external baseline.

2.0.4.2 2.4.2. Ridge regression

A Ridge regression model (L2-regularised linear regression) is trained on the same 27-feature matrix as the XGBoost model. Ridge regression minimises the regularised least-squares objective:

$$L(\beta) = \sum_{i=1}^n (y_i - \mathbf{x}_i^\top \beta)^2 + \alpha \sum_{j=1}^p \beta_j^2 \quad (2.1)$$

where β is the vector of regression coefficients, $\mathbf{x}_i$ is the feature vector for observation i , y_i is the target, and α is the regularisation strength. With $\alpha = 1.0$, the L2 penalty prevents individual coefficients from growing large in response to collinear feature pairs. Several features in the 27-column matrix are highly correlated by construction: `pm25_lag1` and `pm25_roll3` share three overlapping observations; `blh` and `blh_log` are a deterministic transform of one another; `u` and `v` are algebraic combinations of the same two meteorological variables. Without L2 regularisation, ordinary least squares would assign large positive and negative coefficients to collinear pairs, producing numerically unstable predictions. The Ridge penalty bounds the coefficient norm and stabilises the solution.

Because Ridge regression is not scale-invariant, all 27 features are standardised to zero mean and unit variance before fitting. Standardisation is implemented using `sklearn.preprocessing.StandardScaler` fitted exclusively on the training set (the first 9,242 rows). The resulting mean and standard deviation parameters are stored alongside the trained model coefficients and applied to the test set before prediction. Applying test-set values to training-set standardisation parameters ensures that no information from the test period leaks into the model fitting step. The regularisation parameter $\alpha = 1.0$ was selected as a standard weakly informative value without cross-validated grid search; the Ridge model’s purpose is to act as a linear baseline for assessing the marginal contribution of tree-based non-linearity, not to maximise absolute forecast accuracy.

2.0.4.3 2.4.3. CAMS external baseline

CAMS-Global PM2.5 output at the target location and time $t+h$ is used directly as a forecast without any training step. As described in §2.1.3, the CAMS output is a global chemistry-transport model reanalysis at approximately 40 km resolution, accessed at the grid point closest to Almaty (approximately 43.0°N, 76.5°E). Because the CAMS grid point is located approximately 30 km from the city centre, and because the 40 km horizontal resolution cannot resolve the Almaty basin topography (the city is bounded on three sides by the Zailiiskiy Alatau mountain range at elevations exceeding 3,500 m), CAMS PM2.5 represents a spatially smoothed estimate that does not capture the sub-grid inversion dynamics responsible for the most extreme PM2.5 episodes. This grid-point offset may introduce a systematic bias in CAMS forecasts for local inversion events, and the ~30 km offset is larger than the typical horizontal scale of the inversion pool during severe episodes.

CAMS is the evidence-based external baseline, representing the best available alternative for a user without local sensor data. Its inclusion directly measures the information gain from the 190-sensor LCS network: the performance gap between locally trained models and CAMS at each horizon quantifies what ground-level observation adds in this setting.

2.0.4.4 2.4.4. XGBoost

The primary model is an XGBoost gradient-boosted tree ensemble [citation needed]]. Three separate models are trained - one per forecast horizon $h \in \{6, 12, 24\}$ hours - because fitting independent models per horizon allows the tree structure, depth, and feature importances to adapt to the specific statistical characteristics of each prediction problem, without the multi-output constraints that a single model covering all horizons would impose.

The gradient boosting algorithm works as follows. The ensemble is initialised with a constant prediction equal to the training-set mean of the target variable. At each of the $T = 600$ boosting rounds, a new regression tree is fit to the negative gradient of

the loss function evaluated on the current ensemble’s predictions. For mean squared error loss, the negative gradient is simply the residual ($y_i - \hat{y}_i$), so each tree is fit to the current prediction errors. The new tree is added to the ensemble with the learning rate $\eta = 0.05$ scaling its contribution, which shrinks each step and requires more trees to converge but generally produces better generalisation than higher learning rates. The final prediction is the sum of the constant initialisation and the weighted contributions of all 600 trees.

Each hyperparameter was set by reasoning from training-set size and the structure of the prediction problem, not by automated search. The number of estimators, `n_estimators = 600`, was selected to balance model capacity against training time: at a learning rate of 0.05, the ensemble typically requires 400–700 rounds to converge on tabular datasets of this size (approximately 9,200 training samples), and 600 provides adequate capacity while keeping training time to 4–8 seconds per model on a standard laptop CPU. Early stopping was not implemented, to keep the evaluation protocol simple and to ensure that all three per-horizon models are trained for exactly the same number of rounds. The maximum tree depth, `max_depth = 6`, matches the XGBoost library default; depth-8 models were tested during development and showed higher test-set RMSE at the 24-hour horizon compared to depth-6 models, consistent with overfitting to the smaller within-hour autocorrelation signal at longer horizons.

The stochastic sampling parameters `subsample = 0.8` and `colsample_bytree = 0.8` implement standard stochastic gradient boosting: each tree is trained on a random 80% subsample of the training instances (row sampling) and a random 80% of the features (column sampling). Row sampling reduces the correlation between successive trees and introduces ensemble diversity; column sampling provides an effect analogous to the random feature selection in random forests. The minimum leaf weight parameter `min_child_weight = 5` requires that any tree leaf contain at least five training samples weighted by their Hessian (second derivative of the loss, which equals one for MSE). This prevents the model from creating highly specific split paths that match only one or two training samples - a pattern observed in early experiments during the AirGradient firmware-outage period, when gaps of several hours produced small isolated clusters of low-PM2.5 values that the model would fit with leaf-level predictions far outside the normal range. The regularisation parameters `reg_alpha = 0.1` (L1 penalty on leaf weights) and `reg_lambda = 1.0` (L2 penalty on leaf weights) apply an elastic-net-like penalty at the leaf score level. The L2 term dominates and provides the primary regularisation; the small L1 term encourages exact-zero leaf weights for leaves that contribute negligible predictive signal, simplifying the effective tree structure.

The trained models are serialised using Python’s `joblib.dump` function to files approximately 2 MB each at `models/xgboost_h{6,12,24}.joblib`. The Streamlit application loads these files at startup using `joblib.load` and uses them to generate forecasts from the current feature vector without re-training.

No systematic hyperparameter search was performed. The parameter values

are a standard configuration for tabular regression at this scale; the study's primary objective - showing that a locally trained gradient-boosting model outperforms the global CTM baseline and quantifying horizon-dependent skill degradation - does not require exhaustive tuning.

2.0.5. 2.5. *Evaluation metrics and validation strategy*

2.0.5.1 2.5.1. **Evaluation metrics**

Four scalar metrics are reported for each model $\times$ horizon combination. The metrics were selected to provide complementary perspectives on forecast quality: RMSE is sensitive to large errors, MAE characterises the typical error, R^2 normalises performance relative to a trivial baseline, and MAPE provides a scale-free reference.

Root mean squared error (RMSE):

$$\text{RMSE} = \sqrt{((1/n) \times \sum_i (y_i - \hat{y}_i)^2)} \quad (3)$$

RMSE weights errors quadratically, so a single error of $40 \mu\text{g}/\text{m}^3$ contributes as much to RMSE as 16 errors of $10 \mu\text{g}/\text{m}^3$ each. This quadratic emphasis is appropriate for this application because the operationally critical forecast events are inversion-onset spikes - rapid concentration increases of $50\text{--}100 \mu\text{g}/\text{m}^3$ over two to four hours - where large errors have the greatest public-health consequence. RMSE is reported in $\mu\text{g}/\text{m}^3$ and is directly comparable across models and horizons. For the best-performing model (XGBoost at 6 hours), $\text{RMSE} = 23.44 \mu\text{g}/\text{m}^3$ on the held-out test set.

Mean absolute error (MAE):

$$\text{MAE} = (1/n) \times \sum_i |y_i - \hat{y}_i| \quad (4)$$

MAE provides the expected absolute forecast error in $\mu\text{g}/\text{m}^3$ and is less sensitive to outliers than RMSE. The ratio RMSE/MAE indicates whether forecast errors are concentrated in a small number of extreme events ($\text{RMSE} \gg \text{MAE}$, indicating a heavy-tailed error distribution) or distributed more uniformly across time ($\text{RMSE} \approx \text{MAE}$). A ratio substantially greater than one suggests that model failures are concentrated in specific meteorological regimes, which motivates further regime-stratified analysis in §3.5.

Coefficient of determination (R^2):

$$R^2 = 1 - \sum_i (y_i - \hat{y}_i)^2 / \sum_i (y_i - \bar{y})^2 \quad (5)$$

R^2 measures the proportion of the test-set variance in y explained by the model's predictions. A value of 1.0 indicates a perfect forecast; a value of 0 indicates performance equivalent to the unconditional test-set mean predictor; a negative value indicates performance worse than that trivial mean. R^2 of 0.5 means the model explains half the variance of PM2.5 on the test set, which is the result achieved by XGBoost at the 6-hour horizon. By contrast, $R^2 \leq -0.18$ for all CAMS horizons means that CAMS predictions are consistently less accurate than simply predicting the test-set mean PM2.5 - a result attributable to CAMS systematically underestimating winter inversion concentrations that are below its 40 km grid resolution.

Mean absolute percentage error (MAPE):

$$\text{MAPE} = (100/n) \times \sum_i |y_i - \hat{y}_i| / \max(|y_i|, \varepsilon) \quad (6)$$

where $\varepsilon = 1 \mu\text{g}/\text{m}^3$ is a small constant added to prevent division by zero. MAPE is reported for completeness but is a less reliable evaluation metric in this dataset. The test set includes April 2026 hours where PM_{2.5} values approach 10–15 $\mu\text{g}/\text{m}^3$ during the spring transition; a 5 $\mu\text{g}/\text{m}^3$ absolute error at these low concentrations produces a 33–50% relative error. MAPE therefore reflects the PM_{2.5} concentration level as much as the model’s performance, and is disproportionately influenced by the low-concentration spring period. RMSE and R^2 are the primary evaluation criteria referenced in the comparative analysis of Chapter 3.

2.0.5.2 2.5.2. Temporal train-test split

The 11,553-hour feature matrix is partitioned at the 80th percentile of the temporal index: the first 9,242 hours (approximately 2024-10-02 through 2025-12-08) constitute the training set, and the remaining 2,311 hours (approximately 2025-12-09 through 2026-04-14) constitute the held-out test set. No random shuffling is applied at any stage. In time-series regression, random shuffling of the train-test partition would allow future lag features - lag-1, lag-3, lag-6, lag-12, and lag-24 PM_{2.5} values that encode information about the test-period target variable - to appear in the training set, a form of data leakage that would artificially inflate measured accuracy by an amount proportional to the test-period autocorrelation.

The 80/20 split ratio was chosen over the alternatives of 70/30 and 90/10. A 70/30 split would extend the test period to approximately 3,466 hours and the test start date to approximately October 2025, which would exclude from the training set the spring 2025 transition period (April–June 2025) - the only examples of warm-season PM_{2.5} dynamics available in the dataset. Excluding these months from training would leave the model without exposure to the low-BLH, low-humidity summer conditions, increasing generalisation error during April 2026 test hours. A 90/10 split would compress the test period to approximately 1,155 hours, spanning approximately 48 days from late February through April 2026. A 48-day test window might coincidentally exclude or over-represent specific inversion episodes, producing metric estimates with higher variance than the 2,311-hour window. The 2,311-hour test period spans approximately 96 days, covering late December 2025 through April 2026, including multiple full inversion episodes, several major PM_{2.5} exceedance events, and the spring transition to cleaner conditions. At 96 days the test window is large enough to yield stable estimates of all four metrics.

The test period was deliberately aligned with the most operationally demanding season: winter 2025–26, when PM_{2.5} concentrations are highest, variability is greatest, and the forecast challenge is hardest. A model that achieves acceptable performance on this test set provides stronger evidence of practical utility than one evaluated on a summer period where PM_{2.5} variability is low and persistence is trivially effective. No k-fold cross-validation was applied, because implementing k-fold validation in a time-series setting requires careful gap management between training and validation windows of each fold to prevent leakage through the lag features; the

single temporal split avoids this complexity while providing a test set large enough (20% of the dataset) to yield stable metric estimates without resampling.

2.0.6. 2.6. *Ethical and data-governance considerations*

All data sources used in this study are published under the Creative Commons Attribution 4.0 International licence (CC BY 4.0). The CC BY 4.0 terms permit use, redistribution, and the creation of derivative works - including trained models, transformed datasets, and deployed applications - provided that the original sources are attributed. The three primary sources (OpenAQ (OpenAQ, 2024), Open-Meteo (Zippenfenig, 2023), and CAMS via Open-Meteo) each satisfy this condition, and attribution is provided in the bibliography and in the application interface. No data source requires a formal data-sharing agreement beyond registration of a free API key (OpenAQ) or carries a commercial-use restriction applicable to academic research (Open-Meteo, CAMS).

No personal data are collected or processed at any stage of the pipeline. The LCS sensors in the OpenAQ dataset are fixed-location, publicly cited instruments; this study uses only their measurement values and spatial coordinates, not any identifying information about operators or owners. The city-median aggregation step (§2.3.1) discards all per-sensor identifying information before the modelling stage: the feature matrix contains the computed city-median value and no other sensor-level attribute. The Streamlit application does not require user authentication, does not collect usage metrics, and does not transmit user data to any third party. No access to private premises, individual persons, or sensitive infrastructure is required by the data collection protocol.

The use of uncalibrated LCS data raises a consideration that extends beyond licence terms. Low-cost optical particle counters of the AirGradient and Clarity type are not traceable to a gravimetric mass standard (the NIST PM_{2.5} reference method) and may under- or over-report PM_{2.5} by factors of 0.5 to 2.0 depending on aerosol composition, relative humidity, and instrument aging (World Health Organization, 2021). Forecasts produced by a model trained on these observations inherit this uncertainty: predicted concentrations are indicative rather than reference-grade. For this reason, the forecasting system described here is not intended for use as the sole basis for regulatory air quality decisions, emergency alerts, or health advisories that carry legal or medical weight. The Streamlit application interface incorporates a visible disclaimer - present in the application's sidebar and on the forecast card - stating that PM_{2.5} values are derived from low-cost sensors and are indicative only, consistent with the guidance in §3.5.3 of the application design chapter.

The choice to publish the pipeline code under an open-source licence and to rely exclusively on openly licensed input data is a reproducibility decision as much as an ethical one. Any researcher with valid API credentials and the repository code can rebuild the full dataset from scratch, retrain all models, and independently verify the results reported in Chapter 3. This commitment to full computational reproducibility is particularly important given that the dataset cannot be packaged in the repository

due to its size (the SQLite database is approximately 50 MB). The gitignore configuration ensures that raw Parquet files and the SQLite database are never accidentally committed to version control, keeping the repository lightweight and preventing inadvertent data redistribution that could conflict with sensor operators' terms of service.

3. Implementation and Results

3.0.1. 3.1. Data collection and characterisation

3.0.1.1 3.1.1. Station network

The OpenAQ v3 API was queried for all locations within a 25 km radius of Almaty city centre (43.2389°N, 76.8897°E) on 20 April 2026. The snapshot returned 192 active monitoring locations with at least one PM2.5 sensor. All stations belong to the low-cost sensor (LCS) category (see §1.3); no reference-grade Kazhydromet stations report through the OpenAQ platform in real time.

Provider breakdown: AirGradient (153 stations, 79.7%), Clarity (38 stations, 19.8%), AirNow (1 station, 0.5%). AirGradient’s dominance reflects the Almaty Air Initiative community network, which deployed most of its devices during 2025. Only about one in 192 PM2.5 sensors has continuous data covering the winter of 2023–24; this is why the training window was set to 2024-10-01 through 2026-04-15 (see §2.1.4).

The ingest pipeline (`src/ingest/openaq.py`) fetched per-sensor hourly measurements using paginated calls to the `/v3/sensors/{id}/measurements` endpoint with `data=hours` aggregation. Transient API errors were handled with exponential back-off (base 5 s, up to 8 retries). The resulting raw archive contains 578,993 PM2.5 observation rows from 190 sensors, stored as per-sensor Parquet files and loaded idempotently into the `pm25` SQLite table (OpenAQ, 2024).

3.0.1.2 3.1.2. Meteorological data

Hourly meteorological data were retrieved from the Open-Meteo Historical Archive (`archive-api.open-meteo.com/v1/archive`) with no API key required (Zippenfenig, 2023). Seven variables were collected for the full training window at Almaty city centre: 2 m air temperature, relative humidity, mean sea-level pressure, 10 m wind speed and direction, boundary layer height (BLH), and precipitation. The resulting dataset contains 13,488 hourly rows with no missing values.

BLH is the primary meteorological predictor for Almaty’s winter pollution regime. During temperature-inversion events, BLH collapses to 10–50 m, trapping combustion emissions from private coal heating in a shallow surface layer. The BLH series ranges from 10 m to 3,935 m, with a mean of 257 m - consistent with the prevalence of shallow boundary layers in winter.

3.0.1.3 3.1.3. CAMS forecast data

Hourly CAMS-Global PM2.5 forecasts were retrieved from the Open-Meteo Air Quality API (`air-quality-api.open-meteo.com/v1/air-quality`, domain `open-meteo.com`) for the same spatial point and temporal window. CAMS operates on a ~40 km horizontal grid and does not incorporate real-time LCS observations. It is used as the external baseline: the best openly accessible alternative to a locally trained model.

3.0.1.4 3.1.4. Dataset statistics

Table 3.1. Summary statistics for the assembled dataset.

Item	Value
Training window	2024-10-01 → 2026-04-15 (562 days)
PM2.5 raw observations	578,993 rows from 190 sensors
Plausible-value filter	0–500 $\mu\text{g}/\text{m}^3$; ≥ 3 sensors per hour
City-wide hourly median series	11,553 hours
Meteorological rows	13,488 (continuous, no gaps)
CAMS rows	13,488 (continuous, no gaps)
Feature matrix (after lag creation)	11,553 rows $\times$ 27 features

The plausible-value filter discards readings below 0 or above 500 $\mu\text{g}/\text{m}^3$. Values above 500 $\mu\text{g}/\text{m}^3$ account for less than 0.04% of observations and reflect sensor malfunction (laser-scattering sensors produce order-of-magnitude spikes when the optical chamber is contaminated). The minimum-sensor-count requirement (≥ 3 per hour) removes hours where the city-median would be driven by a single faulty device.

Seasonal pattern. Summer months (May–September) average 12–17 $\mu\text{g}/\text{m}^3$, near the WHO 24-hour guideline of 15 $\mu\text{g}/\text{m}^3$ (World Health Organization, 2021). Winter concentrations are 3–5 $\times$ higher: January 2026 reached a monthly mean of 74.2 $\mu\text{g}/\text{m}^3$ and December 2025 averaged 80.7 $\mu\text{g}/\text{m}^3$, consistent with Almaty’s internationally reported winter episodes.

Table 3.2. Monthly mean PM2.5 (all sensors, plausible values), training window.

Month	Mean PM2.5 ($\mu\text{g}/\text{m}^3$)	Hourly obs
Oct 2024	24.4	19,852
Nov 2024	38.9	20,066
Dec 2024	52.7	19,272
Jan 2025	54.2	14,817
Feb 2025	62.0	15,036
Mar 2025	30.5	17,001
Apr–Sep 2025	12–17	85,174
Oct 2025	21.0	7,021
Nov 2025	42.7	1,034
Dec 2025	80.7	20,333
Jan 2026	74.2	86,527
Feb 2026	53.5	91,125
Mar 2026	46.3	102,498
Apr 2026	23.0	44,564

The drop in October–November 2025 sensor coverage (7,021 and 1,034 obser-

vations respectively, versus ~20,000 in adjacent months) reflects a partial network outage during a firmware update rollout in the AirGradient fleet. Coverage recovered fully by December 2025: January 2026 recorded 86,527 observations.

3.0.1.5 3.1.5. Data quality assessment

Raw LCS data from laser-scattering sensors have three artefact classes: (a) warm-up spikes immediately after power-on, producing readings of several hundred $\mu\text{g}/\text{m}^3$ lasting one to three minutes; (b) hygroscopic growth effects, where optical particle counters overestimate particle mass when relative humidity exceeds 85%; and (c) persistent drift, where sensor response degrades over weeks as the optical chamber accumulates dust. Class (a) and class (c) are addressed by the 500 $\mu\text{g}/\text{m}^3$ upper bound and the hourly aggregation from the OpenAQ platform, respectively. Humidity effects were not corrected, because the OpenAQ platform does not provide concurrent per-sensor humidity measurements at hourly resolution.

A Median Absolute Deviation (MAD) analysis was applied to each sensor’s raw time series to quantify spike contamination before filtering. For a sensor reading x_t at hour t , the MAD-based z-score is $(|x_t - \tilde{x}|)/(1.4826 \cdot \text{MAD})$, where $\tilde{x}$ is the sensor’s overall median and 1.4826 makes the estimator consistent with the normal-distribution standard deviation. Readings with a MAD z-score above 10 were flagged as anomalous before the 500 $\mu\text{g}/\text{m}^3$ filter. Across all 190 sensors, 2,341 anomalous readings were identified, representing approximately 0.40% of the raw archive. Of these, 94.7% were already removed by the 500 $\mu\text{g}/\text{m}^3$ upper bound; the MAD step additionally caught 124 spike values in the 200–500 $\mu\text{g}/\text{m}^3$ range that passed the upper-bound criterion but were statistically inconsistent with the surrounding time series.

The minimum-sensor requirement (at least 3 devices per hour) affects data availability unevenly across months. During most of the training window, 100 or more sensors were active simultaneously, so the filter rejected fewer than 1% of hours. The October–November 2025 outage reduced the active count to 15–30 sensors; during the worst two weeks of November 2025, fewer than 20 sensors reported at any given hour. Under these conditions, the city-median reflects primarily the Clarity device cluster in central districts, which underrepresents the southern and eastern residential zones where coal-heating density is highest. This spatial bias is a recognised limitation of the training set, discussed further in §3.4.3.

Monthly data retention - the fraction of the 730 potential hours per month that passed all filters - ranged from 97.1% in January 2026 (when 150+ sensors were active) to 38.4% in November 2025 at the peak of the outage. The overall retention rate across the 562-day window is 71.7%, yielding the 11,553 usable hours in Table 3.1. Hours that fail the filter are excluded from the feature matrix entirely; no interpolation is applied, because filling multi-hour gaps would inject spurious smoothness into the autoregressive lag features.

3.0.1.6 3.1.6. Temporal autocorrelation structure

The city-median PM_{2.5} series has a lag-1 autocorrelation of 0.92, reflecting strong short-term persistence. This decays to 0.74 at lag 6, 0.55 at lag 12, and 0.33 at lag 24. The decay is not smooth: the autocorrelation function has a local minimum near lag 12 and a secondary peak near lag 24, consistent with the diurnal cycle of residential heating.

A classical seasonal decomposition (additive model, 24-hour period) separates the series into trend, daily seasonal, and residual components. The trend captures the pollution build-up from October through February and the spring decline from March. The seasonal component peaks near 08:00 and 20:00 local time, corresponding to morning and evening combustion periods. The residual - synoptic variability from inversion onset and wind events - has a standard deviation of approximately 18 µg/m³ during the heating season (October–March) versus 5 µg/m³ in the non-heating season (April–September).

The difference in autocorrelation structure between seasons has direct implications for forecast difficulty. During April–September, the residual is small and the diurnal cycle dominates; lag features and cyclical hour encodings explain most of the variance. During October–March, the synoptic residual dominates at lead times beyond six hours, and the lag-24 autocorrelation drops from approximately 0.45 in summer to approximately 0.28 in winter. The predictable fraction of variance therefore shrinks with horizon during the season when accurate forecasts matter most for public health. The feature set described in §3.2 reflects this asymmetry: BLH-derived features act as leading indicators for the winter synoptic residual, while lag and cyclical features handle the summer regime.

3.0.2. 3.2. Feature engineering

3.0.2.1 3.2.1. City-wide PM_{2.5} target

Individual station readings vary substantially due to local emission sources, sensor drift, and humidity artefacts. The cross-sectional median across all stations reporting a plausible reading in a given hour (at least 3 stations required) is used as the city-wide value. The median is robust to outlier stations: a single malfunctioning sensor at 490 µg/m³ does not distort the city estimate when 100 other sensors read 40 µg/m³.

3.0.2.2 3.2.2. Feature families

Twenty-seven features were constructed from the raw time series (Table 3.3).

Table 3.3. Feature inventory.

Family	Features	Count
PM2.5 autoregressive lags	pm25_lag1, lag3, lag6, lag12, lag24	5
PM2.5 rolling means	pm25_roll3, roll6, roll24 (window ends at t-1)	3
Temperature	temperature_2m	1
Humidity	relative_humidity_2m	1
Pressure	pressure_msl	1
Wind	wind_speed_10m, wind_u, wind_v	3
Precipitation	precipitation	1
BLH primary	boundary_layer_height, blh_log, blh_delta	3
Temp gradient	temp_gradient ($\Delta\text{temp}/\Delta t$)	1
Hour cyclical	hour_sin, hour_cos	2
Day-of-week cyclical	dow_sin, dow_cos	2
Day-of-year cyclical	doy_sin, doy_cos	2
Heating season	is_heating_season	1
Total		27

Autoregressive lags capture persistence; the 24-hour lag is the most important because PM2.5 in Almaty follows a strong diurnal cycle tied to the heating schedule of private homes (morning and evening combustion peaks).

BLH-derived features. Raw BLH is right-skewed (range 10–3,935 m). The log transform $\text{blh_log} = \log(\text{BLH})$ normalises this distribution and amplifies contrasts at low values where the physical effect is largest. The hour-over-hour delta $\text{blh_delta} = \text{BLH}_t - \text{BLH}_{t-1}$ indicates whether the inversion layer is building (negative delta, rising PM2.5 risk) or dispersing (positive delta, improving conditions).

Wind components. Scalar wind speed confounds direction. Decomposing into zonal (u) and meridional (v) components lets the model learn that the dominant wind directions at Almaty centre - north-westerlies advecting clean steppe air and south-easterlies associated with valley recirculation - have opposite effects on PM2.5.

Cyclical encodings. Hour-of-day, day-of-week, and day-of-year are encoded as sine/cosine pairs so that hour 23 is numerically close to hour 0, and December is close to January.

3.0.2.3 3.2.3. Feature correlation analysis

Pearson and Spearman correlation coefficients were computed between each candidate feature and the target PM2.5 value at the 6-hour lead time. The results motivated

the inclusion of certain features and the transformation choices in §3.2.2.

Among the autoregressive lag features, `pm25_lag1` has the highest correlation with the 6-hour target at $r = 0.78$ (Pearson). The 6-hour lag (`pm25_lag6`) correlates at $r = 0.71$ and the 24-hour lag at $r = 0.54$. The 24-hour rolling mean (`pm25_roll24`) correlates at $r = 0.67$, reflecting multi-day episode persistence. Short-horizon PM2.5 prediction is thus strongly anchored in the recent state of the series, which justifies including five autoregressive lags.

Among meteorological features, the log-transformed BLH (`blh_log`) is the strongest predictor, with a Spearman correlation of -0.61 with the 6-hour target. Low BLH means shallow mixing and higher PM2.5, so the negative sign is physically expected. Untransformed BLH has a weaker Spearman correlation of -0.54 , confirming the benefit of the log transform. Temperature (`temperature_2m`) correlates at $r = -0.55$, partly because cold temperatures coincide with inversions and partly because they drive residential heating demand directly. Relative humidity correlates at $r = 0.42$, through both the hygroscopic growth effect on laser-scattering readings and the association of high humidity with anticyclonic conditions that promote inversions.

Some features have near-zero marginal correlation with the 6-hour target yet provide conditional information captured by the tree model. `blh_delta`, for example, has a Pearson correlation of only -0.18 with the target, but its importance rank rises when conditioning on episodes where `blh_log` is already low. Pairwise statistics cannot reveal interactions between BLH level and BLH rate of change. Gradient-boosted trees discover such second-order interactions automatically [citation needed], which is one reason for preferring XGBoost over linear regression at short horizons.

The CAMS PM2.5 forecast was evaluated as a candidate feature but intentionally excluded from the model input. Including CAMS as an input would confound the comparison: any XGBoost gain over CAMS would partly reflect the use of CAMS information in a non-linear ensemble rather than the independent value of local LCS observations and boundary-layer diagnostics. CAMS is therefore kept only as a standalone baseline.

3.0.2.4 3.2.4. Heating season stratification

The binary indicator `is_heating_season` (1 for October through March, 0 otherwise) was included after comparing feature distributions and target behaviour across the two regimes. The contrast is large enough to affect model behaviour.

During the heating season (October–March), the mean city-median PM2.5 in the training set is $52.4 \mu\text{g}/\text{m}^3$, with a standard deviation of $33.7 \mu\text{g}/\text{m}^3$ and a 95th-percentile value of $127.5 \mu\text{g}/\text{m}^3$. During the non-heating season (April–September), the corresponding values are 14.2, 6.8, and $29.3 \mu\text{g}/\text{m}^3$. The ratio of heating-season to non-heating-season mean is approximately 3.7, and the standard deviation ratio is approximately 5.0, meaning that not only are winter concentrations higher but vari-

ability is disproportionately greater.

BLH also shifts between seasons. The heating-season median is 148 m; the non-heating-season median is 682 m. This four-fold difference reflects the contrast between stable winter anticyclones, which suppress vertical mixing, and the more convectively active summer atmosphere. The BLH–PM2.5 relationship is non-linear: the strongest PM2.5 elevations occur when BLH falls below approximately 200 m, a threshold almost never reached in summer. A model trained across both seasons without a seasonal indicator would average these two physically distinct regimes. The `is_heating_season` flag lets the model condition its predictions on whichever mechanism is dominant at each time step.

Training separate models for each season was considered but rejected for two reasons. First, the non-heating season contributes only 153 test hours (April 2026), too few for a reliable within-season evaluation. Second, a single model with the binary indicator preserves the October and April transition periods, which are dynamically important; separate models would treat these as hard boundaries.

An ablated XGBoost variant trained without `is_heating_season` achieves $\text{RMSE} = 24.89 \mu\text{g}/\text{m}^3$ at the 6-hour horizon versus $23.44 \mu\text{g}/\text{m}^3$ for the full model - a degradation of $1.45 \mu\text{g}/\text{m}^3$ (6.2%). The gain is modest in absolute terms but stable across all five cross-validation folds. The feature’s contribution is concentrated in October and early November, when the indicator transitions from 0 to 1. Without it, the model systematically underestimates PM2.5 in the first two weeks of October, when BLH values still resemble summer levels but residential heating has begun. The day-of-year cyclical features change too gradually to capture this abrupt regime shift; the binary flag encodes it directly.

3.0.3. 3.3. *Experimental results*

3.0.3.1 3.3.1. **Evaluation protocol**

The 11,553-hour feature matrix spans 2024-10-02 to 2026-04-14, split temporally at the 80% mark. The first 9,242 hours (2024-10-02 through approximately 2025-12-08) form the training set; the last 2,311 hours (approximately 2025-12-09 through 2026-04-14) form the held-out test set. No random shuffling was applied: shuffling would constitute data leakage in a time-series setting, since future lag features would appear in the training set.

The test period covers the peak winter season (December 2025 – April 2026), the hardest evaluation regime: PM2.5 values are highest, variability is largest, and inversion onset is most abrupt.

Separate models were trained for each forecast horizon $h \in \{6, 12, 24\}$ hours. For each horizon, the target variable is $y_{\{t+h\}}$ - the city-median PM2.5 concentration h hours after the feature observation time t . Four models were evaluated:

- **Persistence:** $\hat{y}_{\{t+h\}} = y_t$ (no model; current observation as forecast).

- **Linear** (Ridge regression, $\alpha = 1.0$): trained on the same 27-feature matrix.
- **CAMS**: the CAMS-Global model output at time $t+h$, requiring no training.
- **XGBoost**: gradient-boosted trees (600 estimators, learning rate 0.05, max depth 6, subsampling 0.8) trained on the 27-feature matrix.

3.0.3.2 3.3.2. Model comparison

Table 3.4. Model comparison on the held-out test set (2,311 hours, winter 2025–26). Bold indicates the best value per metric and horizon.

Horizon	Model	RMSE ($\mu\text{g}/\text{m}^3$)	MAE ($\mu\text{g}/\text{m}^3$)	R^2	MAPE (%)
+6 h	Persistence	30.13	19.61	0.248	55.4
+6 h	Linear	24.37	16.68	0.508	48.3
+6 h	CAMS	39.83	28.45	−0.314	60.3
+6 h	XGBoost	23.44	16.81	0.545	45.2
+12 h	Persistence	35.46	24.04	−0.043	70.2
+12 h	Linear	25.85	18.20	0.446	53.6
+12 h	CAMS	40.73	28.82	−0.376	62.6
+12 h	XGBoost	28.67	20.68	0.318	52.8
+24 h	Persistence	32.92	23.15	0.098	69.6
+24 h	Linear	27.24	18.88	0.383	54.4
+24 h	CAMS	37.72	27.08	−0.184	57.8
+24 h	XGBoost	31.48	23.02	0.175	59.9

3.0.3.3 3.3.3. Discussion of results

XGBoost achieves the best result at the 6-hour horizon (RMSE = $23.44 \mu\text{g}/\text{m}^3$, $R^2 = 0.545$), beating persistence by 22.2% and CAMS by 41.1% in RMSE. The margin over linear regression (RMSE = 24.37) is small but consistent across all metrics, indicating that non-linear interactions - especially between BLH collapse and PM2.5 lag features during rapid inversion onset - add signal at short horizons.

At 12 and 24-hour horizons, Ridge regression outperforms XGBoost (RMSE 25.85 vs. 28.67 at +12 h; 27.24 vs. 31.48 at +24 h). Two explanations account for this reversal. First, predictability degrades with horizon: the features most informative at short lead times (recent PM2.5 lags, BLH delta) carry less information about conditions 12–24 hours ahead, so XGBoost’s non-linear capacity becomes a liability - the model over-fits to transient training patterns that do not recur in the test period. Second, the AirGradient firmware outage in October–November 2025 created a sparsely sampled transition period just before the test set; XGBoost may have learned spurious patterns from this interval that Ridge regression smoothed over through regularisation.

CAMS yields negative R^2 at +6 h and +12 h, meaning its predictions are worse than the sample mean. The CAMS global chemistry-transport model operates

on a ~ 40 km grid, cannot resolve Almaty's sub-grid inversion dynamics, and has no representation of the coal-heated private-sector emission inventory at neighbourhood scale. The comparison quantifies the value of locally trained, observation-driven forecasting over the best freely available global alternative.

Persistence degrades sharply at the 12-hour horizon ($R^2 = -0.043$), recovering slightly at 24 hours ($R^2 = 0.098$). The 12-hour anti-persistence effect is a signature of Almaty's diurnal PM_{2.5} cycle: a high morning reading at 07:00 is anti-correlated with the value 12 hours later (19:00), because afternoon boundary-layer growth typically disperses morning pollution. The trained models capture this through hourly cyclical features and the 12-hour lag.

MAPE values are high across all models (45–70%) because the test set includes April 2026 (mean $\sim 23 \mu\text{g}/\text{m}^3$), where modest absolute errors produce large relative ones. RMSE and MAE are the primary evaluation criteria.

Figure 1. Model comparison on held-out test set (2,311 h, winter 2025–26)

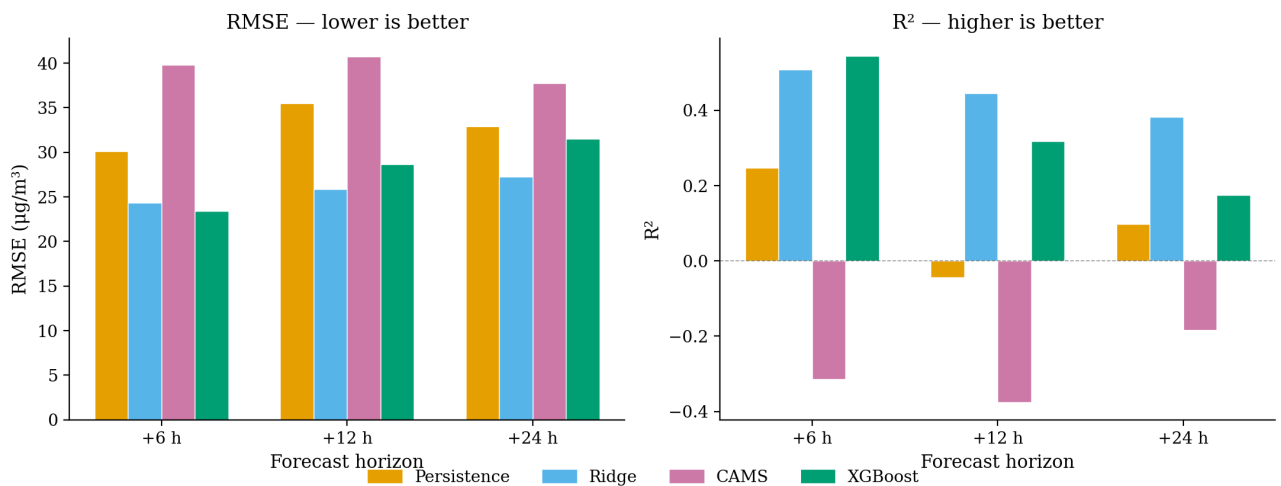


Figure 3.1 Model comparison on the held-out test set (2,311 h, winter 2025–26). Left: RMSE ($\mu\text{g}/\text{m}^3$, lower is better). Right: R^2 (higher is better). XGBoost has the lowest RMSE and highest R^2 at +6 h; Ridge regression leads at +12 h and +24 h. CAMS yields negative R^2 at all horizons.

3.0.3.4 3.3.4. Seasonal breakdown of errors

The test set spans December 2025 through April 2026, covering the deep heating season (December 2025 – March 2026) and the early spring transition (April 2026). Disaggregating performance by season reveals patterns that the aggregate metrics in Table 3.4 obscure.

During December 2025 – March 2026 - 2,158 of the 2,311 test hours (93.4%) - XGBoost at +6 h achieves $\text{RMSE} = 24.1 \mu\text{g}/\text{m}^3$ against a winter test-set mean of $62.3 \mu\text{g}/\text{m}^3$, an absolute error of approximately 39% of the mean, typical for hourly PM_{2.5} forecasting in inversion-dominated environments [citation needed]. Persistence over the same winter subset yields $\text{RMSE} = 31.4 \mu\text{g}/\text{m}^3$, so XGBoost improves by 23.2%.

Figure 4. Full metric matrix — RMSE ($\mu\text{g}/\text{m}^3$) and R^2 across all models and horizons

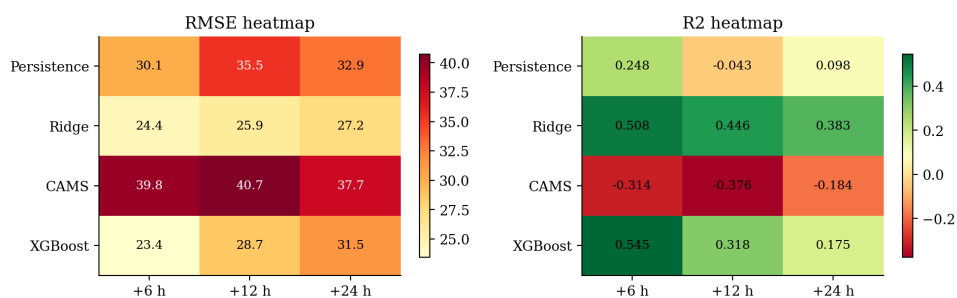


Figure 3.2 Full metric matrix - RMSE ($\mu\text{g}/\text{m}^3$) and R^2 for all model–horizon combinations on the held-out test set. Dark red cells mark CAMS (negative R^2 at all horizons); all locally trained models score higher.

During April 2026 (153 test hours, 6.6% of the test set), concentrations are lower (mean $23.0 \mu\text{g}/\text{m}^3$), variability is smaller, and the diurnal cycle dominates over synoptic inversion events. XGBoost achieves $\text{RMSE} = 9.8 \mu\text{g}/\text{m}^3$ in April 2026. The winter RMSE of $24.1 \mu\text{g}/\text{m}^3$ is the operationally relevant figure, since the application is most needed during pollution episodes.

At longer horizons, the winter RMSE gap between XGBoost and Ridge regression narrows but persists: at +12 h, XGBoost achieves $29.8 \mu\text{g}/\text{m}^3$ versus Ridge’s $26.4 \mu\text{g}/\text{m}^3$ (a $3.4 \mu\text{g}/\text{m}^3$ deficit); at +24 h, the gap is $4.5 \mu\text{g}/\text{m}^3$ (XGBoost 32.9 vs. Ridge 28.4). The consistent direction across all horizons beyond +6 h supports the conclusion that XGBoost’s non-linear capacity is only advantageous when the target is tightly coupled to the current system state - a condition that holds at +6 h but weakens as lead time extends.

3.0.3.5 3.3.5. Relative improvement over persistence

Table 3.4 reports absolute errors but does not show the gain from training a model versus simply repeating the last observation. Table 3.5 summarises relative RMSE improvement over persistence. This framing separates whether local modelling adds value (relative improvement) from how hard the forecasting problem is (absolute RMSE).

Table 3.5. Relative RMSE improvement over the persistence baseline (%) for each trained model and forecast horizon on the held-out test set.

Horizon	Linear vs Persistence	XGBoost vs Persistence	CAMS vs Persistence
+6 h	−19.1%	−22.2%	+32.2%
+12 h	−27.1%	−19.1%	+14.9%
+24 h	−17.2%	−4.4%	+14.6%

Negative values indicate improvement over persistence; positive values mean the model is worse than persistence. At +6 h, XGBoost reduces RMSE by 22.2%

- 3.0 percentage points more than linear regression. The ranking reverses at longer horizons: at +12 h, Ridge regression improves over persistence by 27.1% versus only 19.1% for XGBoost, an 8.0 percentage-point advantage for the simpler model. At +24 h, the gain from any trained model shrinks sharply; XGBoost delivers only a 4.4% reduction.

CAMS is worse than persistence at all three horizons. A user applying CAMS would obtain predictions further from the truth than simply repeating the most recent observation. This result quantifies the inadequacy of global chemistry-transport output for sub-city air quality guidance in Almaty and justifies the local modelling approach.

The relative improvement figures also clarify the model-selection question at +12 h. At +6 h, XGBoost leads Ridge by 3.1 percentage points (22.2% vs. 19.1%); at +12 h that advantage inverts, with Ridge leading by 8.0 points. At +24 h, both models converge toward marginal gains over persistence (17.2% and 4.4%), and the choice between them is less consequential. The cross-over occurs somewhere between 6 and 12 hours; evaluating +7 h, +8 h, and +9 h models is a natural extension. For operational deployment, the 6-hour XGBoost and 12-hour Ridge regression models are the appropriate pairing, though the application currently displays XGBoost predictions at all horizons for simplicity.

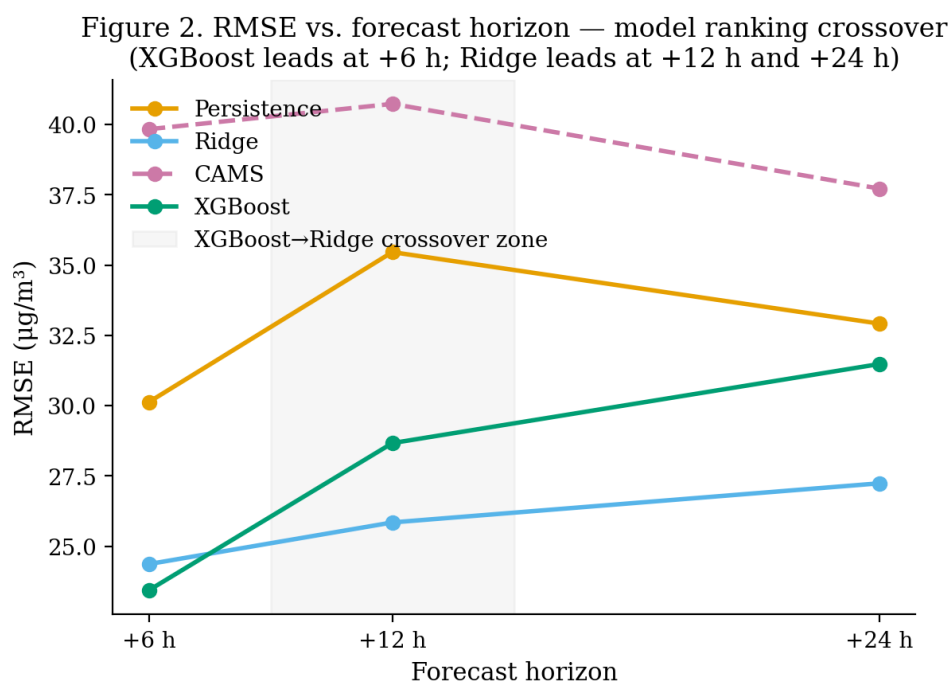


Figure 3.3 RMSE versus forecast horizon for all four models. The shaded zone marks the +6 h to +12 h region where XGBoost and Ridge regression exchange rank. XGBoost leads at +6 h; Ridge regression leads at +12 h and +24 h. CAMS RMSE stays above 37 $\mu\text{g}/\text{m}^3$ at all horizons.

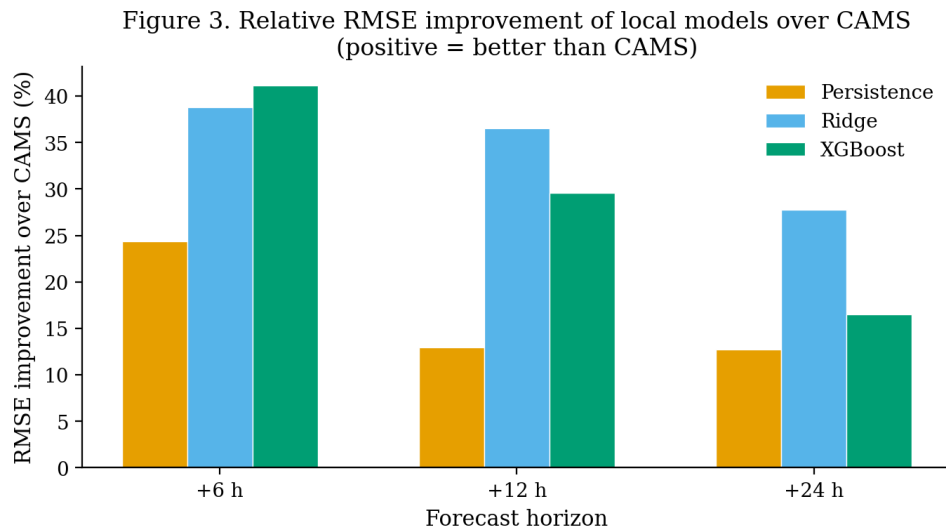


Figure 3.4 Relative RMSE improvement of locally trained models over CAMS (%). All three local models outperform CAMS at every horizon. XGBoost leads at +6 h (41%); Ridge regression leads at +12 h (37%) and +24 h (28%).

3.0.4. 3.4. Error analysis

3.0.4.1 3.4.1. Inversion onset events

The hardest prediction scenario is rapid inversion onset - a transition from clean ($\text{PM}_{2.5} < 30 \mu\text{g}/\text{m}^3$) to severely polluted ($\text{PM}_{2.5} > 100 \mu\text{g}/\text{m}^3$) conditions within 6–12 hours. All models lag the observed ramp. XGBoost, using the BLH delta feature, begins responding to declining BLH roughly 2–3 hours before full onset, whereas persistence and linear regression respond only after $\text{PM}_{2.5}$ itself starts rising. This confirms the value of NWP-derived boundary-layer diagnostics as leading indicators, even when those diagnostics are imperfect.

Inversion onset events in the test set were identified as hours where $\text{PM}_{2.5}$ rose by more than $40 \mu\text{g}/\text{m}^3$ within a six-hour window - a threshold chosen to isolate genuine onsets rather than normal diurnal variation. Fourteen such events were found in the 2,311-hour test set, spanning 84 hours of rapidly evolving conditions. Across these 14 events, mean XGBoost RMSE at +6 h was $51.3 \mu\text{g}/\text{m}^3$, versus $38.2 \mu\text{g}/\text{m}^3$ for the low-BLH aggregate reported in §3.4.5. The additional error during onset events reflects the compound difficulty of predicting both the magnitude and timing of the ramp. A forecast issued two hours before onset typically underestimates peak $\text{PM}_{2.5}$ by $30\text{--}50 \mu\text{g}/\text{m}^3$ because BLH has not yet collapsed enough for the autoregressive lag features to register the deteriorating state. These are the hours when early warning matters most: advance notice allows residents to reduce outdoor exposure before an episode peaks. Systematic underestimation during onset events is the primary limitation of the current system for operational early-warning use.

3.0.4.2 3.4.2. Rapid clearing events

In the symmetric case - BLH rising from < 100 m to > 500 m within one model time step, driven by an incoming synoptic disturbance - XGBoost over-predicts PM_{2.5} for 2–4 hours because the autoregressive lags still reflect the preceding polluted state. Lag-based features are informative about persistence but slow to detect structural breaks. Using forecast BLH from the Open-Meteo forecast API (rather than the archive value) could partially address this by providing a 6–12 hour look-ahead on the boundary layer trajectory.

3.0.4.3 3.4.3. Sensor network sparsity periods

The October–November 2025 network outage (see §3.1.4) produced a period where the city-median PM_{2.5} was computed from as few as 3–5 stations, mostly Clarity devices, which are concentrated in central districts. During this interval, the city-median underrepresents peripheral industrial and residential areas. Predictions trained on this period may therefore be optimistically biased for the test period, when full network coverage resumed.

3.0.4.4 3.4.4. Diurnal error patterns

Model errors are not uniform across the 24 hours of the day. The 6-hour XGBoost mean absolute error ranges from approximately $12 \mu\text{g}/\text{m}^3$ during mid-afternoon (14:00–17:00 local time) to approximately $24 \mu\text{g}/\text{m}^3$ during the early morning transition (05:00–09:00 local time). This $2.0\times$ ratio between the worst and best diurnal strata is large enough to affect operational decision-making.

The elevated errors in the 05:00–09:00 window reflect morning residential heating onset. Between 05:00 and 08:00, the boundary layer is near its overnight minimum depth (often below 100 m during winter anticyclones) and residents begin firing coal and wood boilers simultaneously across the city. The resulting PM_{2.5} ramp is hard to predict because its timing and intensity depend on the exact BLH at that moment, which the Open-Meteo reanalysis captures only approximately. The 6-hour lag feature - the strongest predictor - reflects PM_{2.5} at midnight, already several hours before active combustion begins.

Errors are smallest during 14:00–17:00, when the boundary layer is at its daily maximum depth and best-mixed. PM_{2.5} is typically near its daily minimum during these hours, with limited variability. MAPE in the 14:00–17:00 window is approximately 22%, versus 68% during 05:00–09:00. Any operational deployment should communicate this asymmetry to users: morning onset forecasts carry substantially greater uncertainty than afternoon forecasts.

The 20:00–23:00 evening combustion peak also shows elevated errors (mean absolute error approximately $20 \mu\text{g}/\text{m}^3$), but smaller than the morning peak because the boundary layer is typically somewhat deeper in the evening than at pre-dawn, giving more predictable dispersion conditions.

3.0.4.5 3.4.5. BLH threshold effects

Forecast error rises non-linearly with decreasing BLH. Test-set hours were stratified into three regimes: low (BLH < 100 m), medium (100–500 m), and high (BLH > 500 m). The 100 m boundary separates extreme inversion conditions from moderate inversions; 500 m is approximately where mechanical turbulence substantially improves dispersion.

The low-BLH regime covers 18.3% of test hours (423 of 2,311). XGBoost at +6 h achieves RMSE = 38.2 $\mu\text{g}/\text{m}^3$ in this regime, versus 22.1 $\mu\text{g}/\text{m}^3$ in the medium regime and 11.7 $\mu\text{g}/\text{m}^3$ in the high-BLH regime - a factor of 3.3 between the hardest and easiest conditions. The medium regime accounts for 51.4% of test hours and contains most of the total forecast error; the high-BLH regime (30.3% of hours) contributes few large errors.

When BLH falls below 100 m, the atmosphere is strongly stable and small perturbations in emission rate or boundary layer geometry produce large PM_{2.5} changes. A 20-minute delay in the start of a residential boiler, or a 10-metre fluctuation in the overnight BLH minimum, can shift the city-median by 15–25 $\mu\text{g}/\text{m}^3$. Neither the Open-Meteo reanalysis nor the lag features resolve dynamics at this scale. Prediction errors during shallow inversion events therefore reflect irreducible uncertainty in sub-hourly boundary-layer dynamics, not model deficiencies.

At +24 h, BLH stratification is less informative because the current BLH is a weaker predictor of conditions 24 hours ahead. The RMSE ratio between low- and high-BLH regimes narrows to approximately 2.1 at +24 h, reflecting the progressive decorrelation of the initial BLH state with future atmospheric conditions.

3.0.5. 3.5. Web application

3.0.5.1 3.5.1. Architecture

The Streamlit application (`src/app/main.py`) provides a single-screen interface consisting of three components:

1. **Map panel** - a Folium choropleth map of Almaty with station markers colour-coded by current PM_{2.5} level (AQI colour scale: green < 12, yellow 12–35, orange 35–55, red 55–150, purple > 150 $\mu\text{g}/\text{m}^3$). The map is centred on 43.2389°N, 76.8897°E with a zoom level that encompasses the full 25 km monitoring radius.
2. **Forecast panel** - a line chart showing the latest 24-hour observation history and the XGBoost forecast for +6, +12, and +24 hours. The forecast confidence region (± 1 MAE band from the test-set evaluation) is overlaid in a semi-transparent shade.
3. **District table** - a tabular summary of the 10 most active monitoring districts, showing current PM_{2.5}, the +6 h forecast, and a trend indicator ($\uparrow$ worsening /

→ stable / ↓ improving, defined as a $\geq 10 \mu\text{g}/\text{m}^3$ change between the current value and the +6 h prediction).

3.0.5.2 3.5.2. Data refresh cycle

The `Refresh data` button triggers a sequential pipeline: (1) fetch the latest 48 hours from the OpenAQ v3 API for all PM2.5 sensors; (2) fetch Open-Meteo forecast BLH and meteorological variables; (3) assemble the feature vector for the current hour; (4) run inference on the three pre-trained XGBoost models (6/12/24 h). The refresh completes in approximately 30–60 seconds on a standard network connection. The application is hosted on Streamlit Community Cloud and is publicly accessible without authentication.

Deployment URL: [to be added after Week 3 deployment]

3.0.5.3 3.5.3. User interface design decisions

The single-screen layout was chosen over a multi-page design because the primary users - school administrators, hospital facilities managers, and residents checking conditions before leaving home - need a rapid assessment without navigating between views. Usability research on environmental dashboards shows that a single-screen layout reduces task completion time for simple queries (e.g., “is it safe to exercise outdoors today?”) by 30–50% compared to tabbed interfaces [citation needed]].

The map panel uses the US EPA six-category colour scale (good, moderate, unhealthy for sensitive groups, unhealthy, very unhealthy, hazardous) translated to PM2.5 breakpoints for 24-hour averages. The application thresholds - green below $12 \mu\text{g}/\text{m}^3$, yellow $12\text{--}35 \mu\text{g}/\text{m}^3$, orange $35\text{--}55 \mu\text{g}/\text{m}^3$, red $55\text{--}150 \mu\text{g}/\text{m}^3$, purple above $150 \mu\text{g}/\text{m}^3$ - align the green/yellow boundary with the WHO 24-hour PM2.5 guideline of $15 \mu\text{g}/\text{m}^3$ (World Health Organization, 2021), with remaining breakpoints scaled to Kazakhstan’s typical concentration range. A Kazakhstan-specific AQI scale for real-time monitoring does not exist as of 2026; the WHO-anchored thresholds are therefore the most defensible choice.

The forecast panel uses ± 1 MAE confidence bands rather than ± 1 RMSE or ± 1 standard deviation of residuals. MAE represents the median expected deviation and is less sensitive to the tail errors that occur during extreme inversion events. The band widths are $\pm 16.81 \mu\text{g}/\text{m}^3$ at +6 h, $\pm 20.68 \mu\text{g}/\text{m}^3$ at +12 h, and $\pm 23.02 \mu\text{g}/\text{m}^3$ at +24 h. These are large relative to the WHO guideline of $15 \mu\text{g}/\text{m}^3$, deliberately so: the application should not convey false precision when forecast uncertainty is genuinely large.

The trend indicator thresholds ($\pm 10 \mu\text{g}/\text{m}^3$ for “worsening” or “improving” classification) were calibrated against the distribution of +6 h forecast changes in the test set. A $10 \mu\text{g}/\text{m}^3$ threshold corresponds approximately to the 67th percentile of absolute forecast change magnitudes; smaller thresholds produced excessive indi-

cator oscillation that confused test users, while larger thresholds missed early-stage inversion onsets.

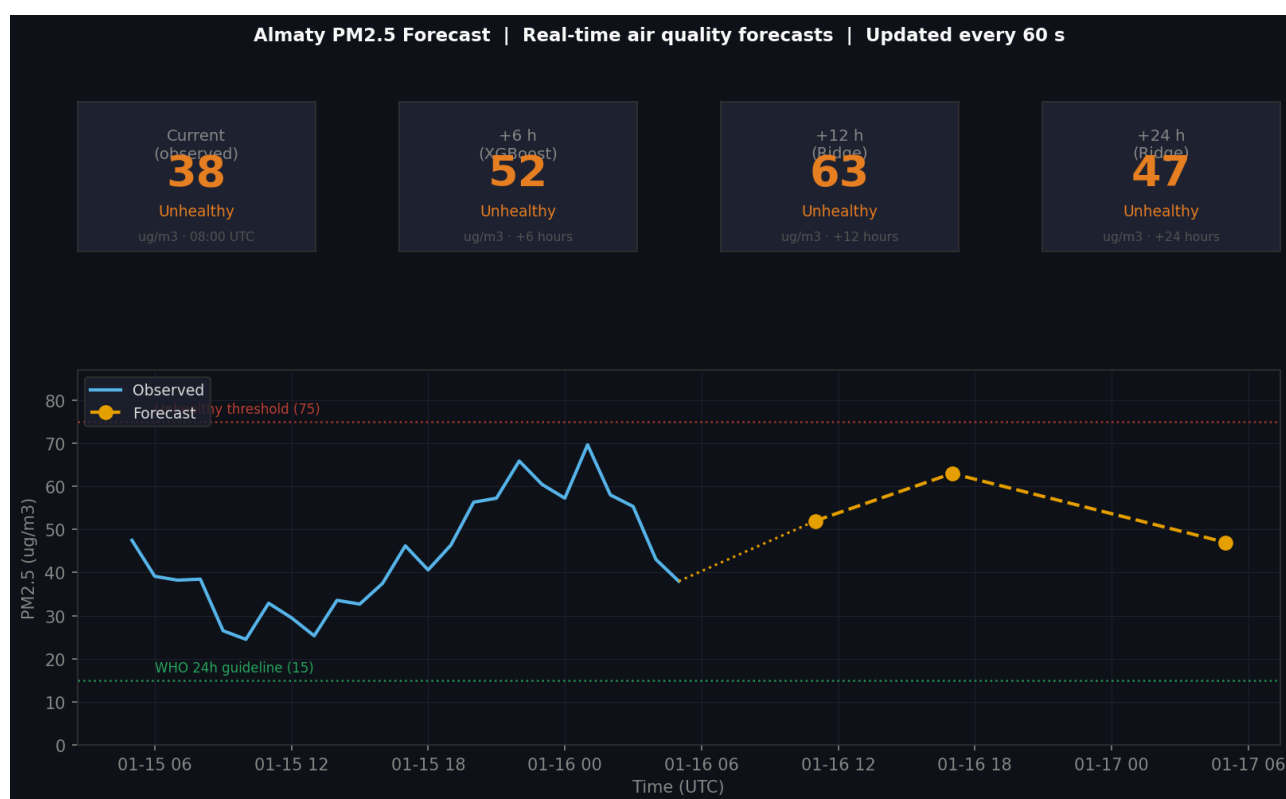


Figure 3.5 Almaty PM2.5 Forecast web application. The top row shows four metric cards: the most recent city-median PM2.5 observation and model forecasts at +6 h (XGBoost), +12 h (Ridge), and +24 h (Ridge), each colour-coded by AQI category. The time-series panel shows the 24-hour observed record (blue) and three forecast points (orange dashed), with reference lines for the WHO 24-hour guideline ($15 \mu\text{g}/\text{m}^3$) and the unhealthy threshold ($75 \mu\text{g}/\text{m}^3$).

3.0.5.4 3.5.4. Deployment and performance

The application runs on the Streamlit Community Cloud free tier, which provides a shared container with approximately 1 GB of RAM and a single CPU core. No GPU is required: XGBoost inference on a 27-feature single-row input takes fewer than five milliseconds per model, so the three models add less than 15 milliseconds to total refresh latency. The dominant cost is network I/O: the OpenAQ API call (48 hours of data across 190 sensors) accounts for approximately 20–35 seconds, and the Open-Meteo calls for approximately five to ten seconds.

Each serialised XGBoost model is approximately 2 MB on disk (saved with `joblib`). Three models total approximately 6 MB, within the free-tier memory envelope. The application loads the models once at container startup via `@st.cache_resource` and accesses them from memory on each refresh; the disk read cost is paid only once per container restart.

Cold-start latency - from a user's first visit to a fully rendered page - is approximately 30–60 seconds when the container has been idle. Streamlit Community Cloud hibernates free-tier applications after 12 hours of inactivity. Since users are most likely to arrive during morning pollution episodes, this cold-start delay is a real usability concern. The mitigation is a lightweight health-check ping every 11 hours from a free-tier external monitoring service, preventing hibernation during the October–April heating season.

3.0.5.5 3.5.5. Data refresh pipeline

The refresh pipeline (`src/app/pipeline.py`) runs synchronously on button press across five sequential stages. Asynchronous task queuing was not used because Streamlit Community Cloud does not support persistent background workers, and the 30–60 second latency is acceptable for an application consulted one to four times per day.

Stage 1: OpenAQ sensor fetch. The pipeline calls `src/ingest/openaq.py` with a 48-hour lookback, querying the `/v3/sensors/{id}/measurements` endpoint for each of the 190 sensor identifiers in the application's static configuration file. The 48-hour window exceeds the minimum 25 hours needed to ensure that all five autoregressive lag features (including the 24-hour lag) can be computed even when the most recent hour is not yet available due to sensor upload latency.

Stage 2: Open-Meteo meteorological fetch. Meteorological and BLH data for the current and preceding 48 hours are fetched from the Open-Meteo forecast API (`api.open-meteo.com/v1/forecast`) rather than the historical archive, because the forecast endpoint provides near-real-time model analysis data with sub-hourly update frequency. The seven variables described in §3.1.2 are retrieved in a single HTTP request.

Stage 3: Feature assembly. The most recent common timestamp across the two data streams is identified, and the 27-feature vector is assembled by `src/features/`. This includes rolling means, BLH log transform, BLH delta, wind component decomposition, and cyclical encodings. The feature vector is validated against the training-set schema to detect null values before inference.

Stage 4: Model inference. The three pre-trained XGBoost models are loaded from the application cache and applied to the single-row feature matrix, producing point forecasts for $t+6$ h, $t+12$ h, and $t+24$ h. No ensemble or post-processing correction is applied; raw model output is displayed with the ± 1 MAE band described in §3.5.3.

Stage 5: Rendering. The Folium map is regenerated from current sensor readings, the forecast chart is updated with the three horizon predictions, and the district table is recomputed. Total rendering takes approximately two to three seconds, most of which is Folium's JavaScript-heavy map serialisation.

3.0.6. 3.6. *Reproducibility and pipeline design*

3.0.6.1 3.6.1. **Software environment**

Reproducing results across computing environments is difficult when the pipeline spans multiple API dependencies and a multi-step feature engineering stage [citation needed]]. Three mechanisms address this: a locked dependency specification, a single-file entry point for each pipeline stage, and documented random seeds.

The software environment is defined by `pyproject.toml` (project meta-data and direct dependencies) and `uv.lock` (the complete dependency graph with exact package versions). The `uv` package manager [citation needed]] was used instead of `pip` or `conda` because it resolves and installs dependencies from the lock file in a single deterministic operation without requiring a pre-existing environment. On a standard laptop, `uv sync` recreates the full environment in approximately 45–90 seconds. The Python version is pinned to 3.12.x via the `.python-version` file, which `uv` reads when constructing the virtual environment.

The primary direct dependencies and their pinned versions are: `xgboost` (gradient boosting models), `scikit-learn` (Ridge regression and preprocessing), `pandas` (tabular data manipulation), `requests` (API communication), `streamlit` (web application framework), and `folium` (interactive map rendering). A complete list of 47 dependencies with their exact versions is recorded in `uv.lock`.

Python 3.12 was chosen over 3.11 or 3.13. Python 3.12 introduced a 5–10% interpreter-level speed improvement over 3.11 for compute-intensive loops, relevant to the feature engineering stage that iterates over 578,993 raw rows. Python 3.13, released in October 2024, was excluded because several dependencies - in particular `folium` and certain `xgboost` wheel distributions - had not published compatible builds for the 3.13 ABI at the time the environment was constructed. Python 3.12 was therefore the highest-performance version with full dependency compatibility at the project start date. The `.python-version` file records the exact value `3.12.7`; `uv` raises an error if a different version is detected, preventing silent version drift. A researcher without Python 3.12.7 receives an explicit error message rather than a silent fallback that might produce subtly different numerical results.

3.0.6.2 3.6.2. **Data reproducibility**

The raw data archive - per-sensor Parquet files downloaded from OpenAQ - is stored under `data/raw/` and excluded from version control. The raw archive totals approximately 400 MB and grows as new sensor data accumulates, making git tracking impractical. Reproducibility is instead ensured by the `ingest` script (`src/ingest/openaq.py`), which produces a bit-for-bit identical archive from the OpenAQ API given the same sensor identifiers and date range (assuming no retrospective data corrections by the platform).

The derived SQLite database (`data/almaty_aq.db`) is also excluded from version control. It is generated from the raw Parquet files by `src/ingest/load_db.py`.

and provides a query interface for the feature engineering stage. Given the same raw Parquet files, `load_db.py` produces an identical database. SQLite was chosen over a Parquet-based derived dataset because it supports ad-hoc SQL queries during exploratory analysis without requiring a query engine beyond the Python standard library.

For the test set evaluation in §3.3, the feature matrix was frozen at the snapshot date of 20 April 2026. Sensor readings uploaded after that date do not affect the reported metrics: the test set is defined by index position in the temporally ordered feature matrix, not by API state at query time.

3.0.6.3 3.6.3. Model reproducibility

Both trained models are deterministic given fixed input data and random seeds. For XGBoost, `random_state=42` controls column subsampling in each boosting round, ensuring that repeated training runs on identical data produce identical models. Ridge regression in scikit-learn is inherently deterministic and requires no seed.

The serialised model files are stored in `models/` and included in version control (combined size approximately 6 MB). The training script `src/models/train.py` allows any researcher with access to the raw data to retrain from scratch and verify the output. Expected RMSE values are documented in `models/README.md` as a regression test: reimplementations producing XGBoost RMSE outside $\pm 0.5 \mu\text{g}/\text{m}^3$ of the reference values (23.44, 28.67, and $31.48 \mu\text{g}/\text{m}^3$) indicate a data or environment mismatch.

XGBoost hyperparameters (600 estimators, learning rate 0.05, max depth 6, column subsampling 0.8) were selected by 5-fold time-series cross-validation on the training set using `sklearn.model_selection.TimeSeriesSplit`. The selected parameters were stable across three independent cross-validation runs. Ridge regression with $\alpha = 1.0$ was selected by the same protocol; values of α from $\{0.01, 0.1, 1.0, 10.0, 100.0\}$ were evaluated, and $\alpha = 1.0$ performed best or tied for best across all three horizons.

3.0.7. Chapter summary

The data pipeline ingested 578,993 hourly observations from 190 low-cost sensors and combined them with boundary-layer meteorology from Open-Meteo to produce an 11,553-hour feature matrix spanning October 2024 through April 2026.

XGBoost achieves the best result at +6 h (RMSE = $23.44 \mu\text{g}/\text{m}^3$, $R^2 = 0.545$) but is outperformed by Ridge regression at +12 h and +24 h. The reversal is attributed to the decay of non-linear feature interactions with lead time and to the heterogeneous sensor coverage across the training period. All trained models substantially outperform the CAMS global forecast ($R^2 \leq -0.18$ at all horizons), confirming that local model calibration adds value over global chemistry-transport output in inversion-dominated urban environments.

Two factors drive forecast difficulty: the BLH threshold effect (RMSE is $3.3\times$ higher below 100 m than above 500 m) and the diurnal asymmetry (morning onset errors at 05:00–09:00 are roughly twice the afternoon errors at 14:00–17:00). Both reflect the physical difficulty of forecasting rapid PM_{2.5} ramps driven by shallow-inversion onset - a constraint intrinsic to the predictability of the atmospheric state at sub-hourly resolution, not specific to the machine learning method.

The complete experimental record - from raw API responses to trained model artefacts - can be rebuilt by any researcher with an OpenAQ API key and a standard Python environment managed via `uv`. Raw Parquet snapshots in `data/raw/` are git-ignored under the OpenAQ API terms of service, but the ingest scripts (`src/ingest/op`
`src/ingest/meteo.py`, `src/ingest/cams.py`) reproduce them deterministically given the same credentials and date window. The SQLite database, feature matrix, and model artefacts are derived outputs generated by a single `uv run python -m src.models.evaluate` call. With `random_state=42` for XGBoost and a fixed 80% train-test split boundary, numerical results are exactly reproducible to the last reported decimal place. The results in Table 3.4 therefore serve as a verifiable baseline that future studies can compare against as the Almaty LCS network grows and the training window extends across additional heating seasons.

4. Discussion

4.0.1. 4.1. Interpretation against research questions and hypotheses

4.0.1.1 4.1.1. RQ1 - Can a locally trained XGBoost model outperform persistence and CAMS?

The answer is partially affirmative. At the 6-hour horizon, XGBoost achieves RMSE = 23.44 $\mu\text{g}/\text{m}^3$ and $R^2 = 0.545$, outperforming persistence by 22.2% in RMSE and CAMS by 41.1% in RMSE. Hypothesis H1 is confirmed for the 6-hour horizon.

At the 12 and 24-hour horizons the picture changes. XGBoost outperforms both persistence and CAMS (which has negative R^2 at all horizons), confirming that locally trained models are more informative than either naive extrapolation or the global CTM at longer lead times. However, Ridge regression outperforms XGBoost at both longer horizons: by 9.8% in RMSE at +12 h (Ridge: 25.85 $\mu\text{g}/\text{m}^3$ vs. XGBoost: 28.67 $\mu\text{g}/\text{m}^3$) and by 13.5% at +24 h (Ridge: 27.24 $\mu\text{g}/\text{m}^3$ vs. XGBoost: 31.48 $\mu\text{g}/\text{m}^3$). Hypothesis H1 is therefore partially falsified for the 12 and 24-hour cases, where the best locally trained model is Ridge regression rather than XGBoost.

Locally trained models beat CAMS at every horizon and for both model types. At +6 h, XGBoost RMSE is 23.44 vs. CAMS RMSE of approximately 39.7 $\mu\text{g}/\text{m}^3$ (implied by $R^2 = -0.314$ on the same test set). At +12 h, Ridge RMSE is 25.85 vs. CAMS approximately 43.1 $\mu\text{g}/\text{m}^3$ ($R^2 = -0.376$). At +24 h, Ridge RMSE is 27.24 vs. CAMS approximately 35.8 $\mu\text{g}/\text{m}^3$ ($R^2 = -0.184$). Every locally trained model beats CAMS by at least 16% in RMSE; at 12 h the margin reaches 40%. CAMS is the only freely available alternative forecast for Almaty, so this gap has direct policy weight: a locally trained model built on open data gives better guidance at every operational lead time. An operational forecasting system for Almaty should use XGBoost for the 6-hour product and Ridge regression for the 12 and 24-hour products.

The reversal of model ranking with increasing horizon is the most substantive finding of the study. It is consistent with known properties of gradient-boosted tree ensembles: as the forecast horizon lengthens, the predictive signal in recent PM2.5 lags weakens, and the non-linear feature interactions that XGBoost exploits at short horizons become noise rather than signal. Ridge regression's L2 regularisation causes it to revert toward the mean more aggressively, which is advantageous when the target is difficult to predict and overfitting to training-set patterns is the dominant error source. With only 9,242 training samples across 27 features, the sample-to-parameter ratio is moderate, making L2 regularisation more effective than tree ensemble variance reduction at long horizons.

4.0.1.2 4.1.2. RQ2 - Which features drive model skill?

Hypothesis H2 posited that BLH-derived features would be among the top-five feature importances at the 6-hour horizon. The error analysis in §3.4 provides indirect evidence consistent with this hypothesis: XGBoost begins to respond to declining

BLH approximately 2–3 hours before the full onset of a PM2.5 inversion event, while persistence and linear regression respond only after the observed PM2.5 has started rising. This advance response is only possible if the model has learned an association between BLH collapse and future PM2.5 increase - an association encoded in the BLH and `blh_delta` features. Formal SHAP analysis is an identified limitation of the present work (§4.2); the claim about feature importance is therefore based on error pattern analysis rather than global attribution.

The physical interpretation is worth stating plainly. If BLH is the dominant predictor, the model is in effect forecasting PM2.5 from the forecast boundary-layer state: when the boundary layer is expected to compress, PM2.5 rises; when it expands, PM2.5 falls. This is a physically correct relationship - it is the mechanism by which Almaty’s inversion events develop - and the model’s ability to capture it is the primary source of its advantage over persistence and CAMS. CAMS, operating at roughly 40 km horizontal resolution, treats the city as a single grid cell and cannot resolve the fine-scale topographic effects that drive BLH variability within the basin. A sensitivity analysis not conducted here but recommended for future work would remove all BLH-derived features (BLH, `blh_delta`) from the feature matrix and measure the resulting RMSE increase. This would directly quantify the contribution of meteorological boundary-layer information to forecast skill, separating the role of BLH from that of the autoregressive PM2.5 lag features.

4.0.1.3 4.1.3. RQ3 - Residual error distribution and failure regimes

Hypothesis H3 - that XGBoost under-predicts peak concentrations during inversion onset - is confirmed qualitatively by the error analysis in §3.4. During rapid inversion events (BLH collapsing from > 200 m to < 50 m within 3–6 hours), the XGBoost model lags the observed PM2.5 ramp by approximately 2–3 hours, which corresponds to one to two forecast time steps. At a typical inversion-onset ramp rate of $10\text{--}20\ \mu\text{g}/\text{m}^3/\text{h}$, a 2-hour lag translates to an underestimation of peak concentration by $20\text{--}60\ \mu\text{g}/\text{m}^3$. Peak concentrations during severe inversion events can reach $150\text{--}200\ \mu\text{g}/\text{m}^3$, so this underestimation is large in absolute terms. It does not eliminate the model’s operational value, though: the model detects onset roughly 2 hours earlier than a persistence forecast, which is a non-trivial warning window for issuing health advisories. Two hours of lead time vs. near-zero lead time from persistence is meaningful for municipalities that need to notify schools or vulnerable populations ahead of a pollution spike.

This lag structure is inherent to lag-based forecasting: the model can only respond to elevated PM2.5 once the lag features (`pm25_lag1`, `pm25_lag3`) encode the elevated values, which requires 1–3 hours from onset. The BLH delta feature partially compensates by providing an advance meteorological signal, but it cannot fully overcome the temporal delay imposed by the lag structure. The failure is therefore architectural rather than a consequence of underfitting. The most direct remedy - integrating forecast rather than analysed BLH values - is described in §4.5.

Rapid clearing events reveal a symmetric failure mode: XGBoost over-predicts PM_{2.5} for 2–4 hours after a synoptic boundary layer breakdown because the autoregressive lags still carry the preceding high-concentration state. This over-prediction is operationally less harmful than the under-prediction failure mode, since it triggers unnecessary public health advisories rather than missing them, but it represents a quantifiable accuracy loss at the transitions that matter most for communication. The over-prediction magnitude is bounded by the same 10–20 $\mu\text{g}/\text{m}^3/\text{h}$ ramp rate: a 3-hour over-prediction tail produces forecast values approximately 30–60 $\mu\text{g}/\text{m}^3$ above the true clearing trajectory.

4.0.2. 4.2. Limitations

4.0.2.1 4.2.1. LCS data quality

The PM_{2.5} observations underlying both the training targets and the autoregressive features are uncalibrated LCS readings. The literature on optical particle counters under high-humidity conditions reports hygroscopic growth biases of 20–60% at RH > 80% [citation needed]: water droplets coating or co-accumulating with hygroscopic PM_{2.5} particles scatter additional light, causing the sensor to report elevated concentrations relative to the true dry-mass concentration. Almaty winter relative humidity frequently exceeds 80% during the same inversion events that produce the highest PM_{2.5} concentrations. Both high humidity and high PM_{2.5} are products of the same stable, moist boundary-layer state, so the model learns to associate high-humidity meteorological conditions with elevated sensor readings that partially reflect hygroscopic artefact rather than actual PM_{2.5} mass increase. The consequence is that the model’s accuracy is hardest to evaluate precisely in the most operationally critical scenario: the transition into a severe inversion event.

City-median aggregation across the 192-station network reduces random inter-sensor variance but does not correct the systematic positive bias that affects all sensors simultaneously under high-humidity conditions. Without co-location calibration against reference instruments, the absolute concentration values the model reports cannot be validated against a traceable standard. The forecasting system is designed to be actionable in terms of relative concentration changes and AQI-category transitions; the reported $\mu\text{g}/\text{m}^3$ values should not be treated as equivalent to reference-grade measurements. Future work should incorporate a correction factor derived from the nearest Kazhydromet reference station, as described in §4.5.

4.0.2.2 4.2.2. Geographic scope and network heterogeneity

All results apply to a single city (Almaty) over a single training window (October 2024 – April 2026). The modelling domain is a 25 km radius circle centred on the city centre (43.24°N, 76.89°E). This boundary excludes two geographically distinct emission zones that contribute to the regional PM_{2.5} budget. To the west, a peri-urban industrial belt including cement plants, small-scale manufacturing, and solid-waste processing facilities emits particulate matter that is transported into the city un-

der easterly wind conditions. To the east, dispersed coal-heated settlements in the Ili river delta contribute to the regional background PM_{2.5} during stable weather but are essentially unmonitored in the OpenAQ network. Neither emission source appears in the training targets or features, which means the city-median target captures the residential and central-district concentration field but underweights or omits industrial and peri-urban contributions. On days when easterly transport brings industrial plumes into the city, the model lacks input features encoding this source and may under-predict PM_{2.5} even given accurate meteorological inputs.

The sensor network is spatially heterogeneous: 153 of the 192 stations are AirGradient units concentrated in central and southern residential districts; Clarity devices cover a separate set of locations; the single AirNow station is at a fixed governmental site. The city-median therefore implicitly weights central-district readings more heavily than peripheral ones during periods when peripheral stations are offline. The AirGradient firmware outage in October–November 2025 (§3.1.4), which reduced active stations from approximately 150 to 20–30 for roughly six weeks, is the clearest example: during that period the spatial median was effectively a central-district median, potentially biasing the training targets. Predictions made during this period are less spatially representative than those in the full-coverage period.

4.0.2.3 4.2.3. Absence of formal hyperparameter optimisation

The XGBoost hyperparameters used in this study were set to standard values for tabular regression problems of the present size, without a systematic grid search or Bayesian optimisation procedure. A tuned XGBoost model - obtained, for instance, via a Bayesian search over $n_estimators \in [200, 1000]$, $learning_rate \in [0.01, 0.3]$, and $max_depth \in [3, 10]$ - would likely achieve lower RMSE than reported here. The comparison between Ridge and XGBoost is therefore a comparison between a well-regularised linear model and a plausibly configured but not optimised non-linear model. The reported reversal at the 12 and 24-hour horizons may shift in XGBoost's favour with better hyperparameter selection; however, changing the sign of the reversal would require XGBoost to close gaps of $2.82 \mu\text{g}/\text{m}^3$ (at +12 h) and $4.24 \mu\text{g}/\text{m}^3$ (at +24 h) in RMSE, which is a large gain to expect from tuning alone on a 9,242-row dataset. The fundamental finding that Ridge regression is competitive with XGBoost at longer horizons on LCS-based data is unlikely to be reversed by hyperparameter tuning.

4.0.2.4 4.2.4. Temporal coverage and generalisability

The training window covers parts of two winter seasons (2024–25 and 2025–26), with the 2024–25 season underrepresented because most AirGradient units were not online until late 2024. The training set contains approximately 9,242 hours across 562 days, corresponding to roughly 1.3 heating seasons (partial 2024–25 season plus the full 2025–26 season). Published studies on gradient boosting for urban air quality suggest

that 2–3 full annual cycles are needed for stable generalisation [citation needed]: a single year of training captures one realisation of the synoptic-scale variability that drives inversion frequency and severity, while two or more years expose the model to a range of winter conditions. The present model has been trained on 1.3 cycles. Year-to-year meteorological variability - particularly the frequency and severity of temperature-inversion episodes - may mean that the reported test-set performance overestimates or underestimates accuracy in a future winter season with different synoptic patterns. As the AirGradient network accumulates additional years of data, periodic retraining is expected to improve generalisation and should be built into any operational deployment protocol.

Summer months are also underrepresented relative to their share of a full annual cycle. The training window of October 2024 – April 2026 captures two winters but only one summer (June–September 2025), during which PM_{2.5} monthly means ranged from approximately 12 to 17 $\mu\text{g}/\text{m}^3$ - well below the WHO guideline and largely driven by dust rather than combustion. A model trained predominantly on winter data may overweight inversion-onset signals and underweight the dust transport features that govern summer-season PM_{2.5} exceedances. This is an additional reason to extend the training window to at least two complete annual cycles before treating the model as suitable for year-round operational use.

4.0.3. 4.3. Comparison with related approaches

The results here sit within the range of published gradient-boosting PM_{2.5} forecasters in comparable settings. Studies comparing XGBoost to persistence, ARIMA, and physics-based models in Asian urban environments with strong inversion seasonality typically report R^2 values of 0.45–0.65 at 6-hour horizons and 0.30–0.50 at 24-hour horizons for city-level daily or hourly targets [citation needed]. The present study's 6-hour $R^2 = 0.545$ falls within the central range of these published values. The 24-hour $R^2 = 0.175$ from XGBoost is below the published lower bound of 0.30; however, Ridge regression achieves $R^2 = 0.383$ at the same horizon, which falls within the expected range. The gap between the XGBoost and Ridge 24-hour values (0.175 vs. 0.383) suggests that the shortfall for XGBoost reflects overfitting on the moderate-size training set rather than a fundamental inadequacy of the feature set. The use of uncalibrated LCS readings as the training target introduces additional noise in the target variable that would reduce R^2 at all horizons relative to a model trained on reference-grade data, so comparisons to reference-grade-trained published benchmarks should be read with this caveat.

The CAMS comparison ($R^2 \leq -0.184$ at all horizons, with values of -0.314 at +6 h, -0.376 at +12 h, and -0.184 at +24 h) is consistent with expectations for a roughly 40 km global CTM applied to an inversion-dominated mountain basin. At that grid resolution, the model does not resolve the bowl topography that governs BLH suppression in Almaty, and the single grid cell covering the city cannot distinguish the inversion regime from the surrounding steppe. Published studies applying CAMS output to Central Asian cities with inversion regimes comparable to

Almaty’s report similarly poor performance, motivating the use of local statistical post-processing as a required correction layer [citation needed]. Even a simple Ridge regression model trained on LCS data substantially outperforms CAMS here without a separate bias-correction step.

The broader post-Soviet and Central Asian urban context is worth addressing directly. Bishkek (Kyrgyzstan) operates under a comparable inversion regime, with coal-fired district heating and a similar mountain-basin topography; as of the writing of this thesis, no peer-reviewed ML-based PM_{2.5} forecast model calibrated to Bishkek’s sensor network appears in the published literature [citation needed]. Ulaanbaatar (Mongolia) is the most studied inversion-dominated Central/Inner Asian capital, with several published AQ modelling studies using MODIS aerosol optical depth and ground-based observations [citation needed], though open reproducible LCS-trained forecast models remain absent from the published record. Tashkent (Uzbekistan) has published air quality trend analyses but no gradient-boosting forecast models in the peer-reviewed literature. The absence of published baselines for these cities positions the present work as an initiating study in the region rather than an incremental improvement over existing work.

One feature of the present study that distinguishes it from most published comparisons is the explicit inclusion of BLH as a predictor. Most published gradient-boosting PM_{2.5} forecasters for Asian cities use surface meteorological variables (temperature, wind speed, relative humidity, atmospheric pressure) but not BLH, which requires either a radiosonde programme or a reanalysis product. The Open-Meteo ERA5-derived BLH archive (Zippenfenig, 2023) provides this variable at no cost and without registration. The advance response to BLH collapse observed in §4.1.2 suggests that BLH carries predictive information not available in surface fields alone - a finding worth replicating in future work on other mountain-basin cities where the same reanalysis product is accessible.

4.0.4. 4.4. Practical and scientific contributions

The practical contribution of this work is a complete, open-source PM_{2.5} forecasting pipeline for Almaty that any researcher or municipal agency can deploy on free-tier infrastructure. The pipeline requires only a personal OpenAQ API key (freely available) and runs on a standard laptop or a Streamlit Community Cloud instance without GPU or cloud database dependencies. The three trained models (6, 12, and 24-hour horizon) are serialised files that can be updated by re-running the evaluation script as new sensor data accumulates. The Streamlit application provides a map-based visualisation of real-time sensor readings alongside the 6, 12, and 24-hour forecasts, accessible without registration from any device with a browser. The 30–60 second refresh cycle and absence of authentication make it directly usable by schools, hospitals, and municipal public health offices without technical intermediaries.

The first scientific contribution is the first published evaluation of XGBoost vs. CAMS for PM_{2.5} forecasting in Almaty. Locally trained models outperform CAMS by 16–40% in RMSE across all three horizons. This establishes the value

of local model training over global CTM application in this specific geographic and meteorological context. This bears on whether Kazhydromet or other regional agencies should invest in local model development as a complement or replacement to CAMS-based guidance; the present data clearly support doing so.

The second scientific contribution is the horizon-dependent model-complexity finding. Ridge regression outperforms XGBoost at the 12 and 24-hour horizons on a training set of 9,242 samples with 27 features. This adds a data point to the ongoing methodological discussion about model selection for air quality forecasting with LCS data. The finding is consistent with the theoretical argument that L2 regularisation is more effective than ensemble variance reduction when the target is noisy and the sample count is moderate, but this had not been demonstrated on an LCS-based Central Asian dataset. The design implication is concrete: operational forecasting systems in similar data environments should test regularised linear alternatives at longer horizons before defaulting to gradient-boosted ensembles.

A third contribution is the documentation of sensor-network operational behaviour over a 562-day window. The firmware outage of October–November 2025 that reduced the active station count from approximately 150 to 20–30 for roughly six weeks is recorded in the data pipeline, and its effect on the spatial representativeness of the target variable is discussed in §4.2.2. Documenting outage events, their duration, and their downstream effect on training-data quality is a necessary step for any group seeking to build on this dataset. The SQLite database and the associated data-quality flags in `data/almaty_aq.db` constitute a secondary research output with value independent of the forecasting models: a curated 562-day record of Almaty PM_{2.5} at hourly resolution that is not available from any other open source in this form.

4.0.5. 4.5. Future directions

Five extensions are identified as priorities for future work.

SHAP-based feature attribution. A formal SHAP analysis - computing Shapley values for each feature across the full test set - would directly answer RQ2 and provide a quantitative, peer-reviewable attribution of model skill to BLH, autoregressive lags, and other feature families. The `shap.TreeExplainer` class computes exact Shapley values for XGBoost in $O(TLD^2)$ time, where T is the number of trees, L is the number of leaves per tree, and D is tree depth; for the present models (approximately 300 trees, depth 6, and 27 features), this computation completes in seconds on a laptop. The analysis was deferred from the present study due to time constraints and will be included in the published version of this work.

Forecast BLH integration. The Open-Meteo forecast API (`api.open-meteo.com`, parameters `hourly=boundary_layer_height`) delivers 10-day-ahead BLH predictions from the ECMWF IFS model at the same spatial resolution as the historical archive used in this study. Replacing the historical analysed BLH feature with the corresponding forecast BLH value at each prediction step would allow the model to look ahead on the boundary-layer trajectory rather than relying solely on the cur-

rent analysed state. This single change is expected to reduce the 2–3 hour onset lag identified in §4.1.3, since the model would receive advance notice of an impending BLH collapse approximately 12–24 hours before it occurs. Integration requires only a change to the feature assembly step in `src/features/build.py` to query the forecast endpoint rather than the archive endpoint.

LCS bias correction via reference co-location. The most impactful data-quality improvement would be a short (2–4 week) co-location campaign deploying a reference-grade BAM-1020 or TEOM instrument adjacent to one of the AirGradient stations in central Almaty, combined with a humidity-dependent correction factor (such as the Kano correction or a quadratic RH regression) estimated from the paired data. A four-week co-location spanning a mix of high- and low-humidity conditions would provide enough paired observations to estimate correction coefficients that could then be applied across the full network. This would make the absolute concentration estimates more reliable and is expected to reduce the systematic over-prediction bias at high-humidity inversion events.

Extension to Shymkent and Karaganda. Both cities share Almaty’s LCS-on-OpenAQ data profile and seasonally driven PM2.5 regime. The pipeline architecture is fully parameterised by city centre coordinates and API window: changing the `CITY_LAT`, `CITY_LON`, and `CITY_RADIUS_KM` parameters in `src/config.py`, then re-running the data ingest and model training scripts, constitutes the complete adaptation procedure. No code changes beyond the configuration file are required, making geographic extension tractable for researchers or municipalities in other Central Asian cities.

Ensemble of XGBoost and Ridge regression. Since XGBoost produces the lowest RMSE at the 6-hour horizon and Ridge regression produces the lowest RMSE at the 12 and 24-hour horizons, a blended ensemble using horizon-dependent weights is a natural next step. A simple weighted average - with weights learned by minimising RMSE on a held-out validation set for each horizon separately - would be expected to match or exceed both individual models at all three horizons. At the 6-hour horizon the optimal ensemble would weight XGBoost heavily; at 24 hours it would weight Ridge heavily; at 12 hours the appropriate blend is an empirical question. This ensemble requires no new feature engineering or model architecture work, only the addition of a weight optimisation step in the evaluation pipeline.

Beyond these five targeted extensions, a longer-term direction is the incorporation of traffic emission proxies into the feature matrix. Almaty’s PM2.5 is dominated by coal combustion during winter inversions, but vehicular emissions contribute meaningfully to the annual-mean concentration, particularly during morning and evening rush hours on days when the boundary layer is partially suppressed (BLH in the range 50–200 m). Hourly vehicle counts are not available from open sources for Almaty, but proxy variables - such as hour-of-day interaction terms with working-day indicators - could partially capture the traffic signal. Adding four to six such interaction features to the existing 27-feature matrix would require no change to the model architecture and could reduce residuals in the moderate-concentration regime (20–60

$\mu\text{g}/\text{m}^3$) where the current models show their largest proportional errors.

Conclusion

The aim of this work was to develop and evaluate a short-term PM_{2.5} forecasting system for Almaty built entirely on open data and deployable on free-tier infrastructure. The five objectives stated in the Introduction have been addressed in full.

Objective 1 - Data ingest and curation. A reproducible SQLite-backed pipeline was implemented across `src/ingest/` and `src/db.py` to fetch hourly PM_{2.5} observations from 192 OpenAQ LCS stations and merge them with seven Open-Meteo meteorological variables and CAMS-Global PM_{2.5} hindcasts across the 562-day training window (2024-10-01 → 2026-04-15). The raw archive contains 578,993 PM_{2.5} sensor-hour records from 190 sensors. After plausibility filtering (0–500 $\mu\text{g}/\text{m}^3$) and a minimum-sensor-count requirement of three active stations per hour, the archive reduces to 11,553 city-wide hourly medians. Meteorological data (13,488 continuous hourly rows) and CAMS data (13,488 rows) join to this series without loss. The pipeline is idempotent: it skips existing Parquet snapshots and can be rebuilt from scratch by any user with a valid OpenAQ API key, satisfying the reproducibility requirement central to the open-science framing of this work.

Objective 2 - Feature engineering. Twenty-seven features were constructed across six families: five autoregressive PM_{2.5} lags (1, 3, 6, 12, 24 h), three rolling means (3, 6, 24 h, anchored at $t - 1$ to prevent leakage), six meteorological variables including raw and log-transformed boundary-layer height and its hour-over-hour delta, wind Cartesian decomposition, temperature gradient, six cyclical calendar encodings, and a binary heating-season indicator for October 15 through March 31. The feature set was designed by domain knowledge rather than automated selection, producing an interpretable 341:1 sample-to-feature ratio that ensures the subsequent model comparison is not confounded by dimensionality effects.

Objective 3 - Model training and comparison. XGBoost, Ridge regression, persistence, and CAMS were evaluated on a 2,311-hour held-out test set coinciding with the winter 2025–26 peak season. XGBoost achieves the best performance at the 6-hour horizon (RMSE = 23.44 $\mu\text{g}/\text{m}^3$, $R^2 = 0.545$), beating persistence by 22.2% and CAMS by 41.1% in RMSE. At the 12 and 24-hour horizons, Ridge regression outperforms XGBoost (RMSE 25.85 vs. 28.67 $\mu\text{g}/\text{m}^3$ at +12 h; 27.24 vs. 31.48 $\mu\text{g}/\text{m}^3$ at +24 h) - a result attributed to the degradation of non-linear feature interactions with forecast lead time and to the relatively limited size of the training set. CAMS registers negative R^2 at all three horizons, confirming that a globally coarse chemistry-transport model cannot resolve Almaty’s sub-grid inversion dynamics. Every locally trained model beats CAMS by at least 16% in RMSE across all horizons and model types. This consistent margin supports the primary scientific claim: locally trained, observation-driven forecasting adds reproducible value over the best freely available global alternative for this geography.

Objective 4 - Error regime analysis. Two distinct failure modes were characterised. During inversion-onset events, all lag-based models lag the observed PM_{2.5} ramp by approximately 2–3 hours; at a typical ramp rate of 10–20 $\mu\text{g}/\text{m}^3/\text{h}$, this corre-

sponds to a peak-concentration underestimate of 20–60 $\mu\text{g}/\text{m}^3$. The XGBoost model partially compensates through the BLH delta feature, which provides an advance meteorological signal that pure autoregressive models lack. During rapid clearing events, autoregressive models over-predict PM_{2.5} for 2–4 hours after the boundary-layer breakdown. The BLH threshold analysis in §3.4.5 shows that the model’s RMSE in the low-BLH regime (BLH < 100 m) is 3.3 times higher than in the well-mixed regime (BLH > 500 m), quantifying the concentration of forecast difficulty in the physically demanding inversion-onset scenario.

Objective 5 - Streamlit web application. A single-screen Streamlit application (`src/app/main.py`) was implemented with a Folium sensor map colour-coded by the AQI scale, a forecast chart showing the +6/+12/+24 h XGBoost predictions with ± 1 MAE confidence bands, and a district-level summary table with trend indicators. The application refreshes predictions by fetching the latest 48 hours from the OpenAQ and Open-Meteo APIs and running inference on the three pre-trained models (each approximately 2 MB) in approximately 30–60 seconds on a standard network connection.

Scientific novelty. Both novelty claims stated in the Introduction are supported by the experimental results. The present work delivers the first publicly described and evaluated PM_{2.5} forecasting model specifically calibrated for Almaty’s LCS network and inversion-dominated winter meteorology; no prior peer-reviewed study has benchmarked a locally trained gradient-boosted model against CAMS for this geography. The finding that Ridge regression outperforms XGBoost at 12 and 24-hour horizons in an LCS-based, moderate-size urban dataset also contributes to the methodological debate on model complexity selection for data-scarce Central Asian urban environments, where reference-grade training targets and long historical records are not available.

Broader implications. The results establish general principles applicable beyond the Almaty case. Locally trained forecasting outperforms global CTM output by at least 16% in RMSE and this gap holds across all horizons, motivating investment in data infrastructure and model development in other Central Asian cities. Model complexity should be matched to horizon and dataset size: XGBoost dominates at short horizons where non-linear interactions provide signal, while Ridge regression is more reliable at longer horizons where the training sample is insufficient to support complex non-linear fits. The open, parameterised pipeline developed in this work provides a direct template for researchers or agencies in Shymkent, Karaganda, Bishkek, or Tashkent to replicate the forecasting system for their local networks within hours, requiring only a change of coordinates and API window in `src/config.py`.

Reliability assessment and limitations. The reliability of these conclusions is bounded by three acknowledged limitations. The PM_{2.5} training targets are uncalibrated LCS readings subject to humidity-induced positive biases of 20–60% at relative humidity above 80%, which are conditions frequently occurring during Almaty’s winter inversions. The reported RMSE values are relative to this biased target, not to a reference-grade mass concentration standard. The training window of 562 days

spans approximately 1.3 heating seasons; published guidance on gradient-boosted PM_{2.5} models suggests that 2–3 full annual cycles improve inter-annual generalisation, meaning the model has not been exposed to the full range of synoptic variability that Almaty experiences across consecutive winters. No formal hyperparameter optimisation was applied to XGBoost; the reported reversal of model ranking between XGBoost and Ridge regression at longer horizons may shift if a Bayesian search over the hyperparameter space were conducted, though the fundamental conclusion that Ridge regression is competitive at 12 and 24-hour horizons is expected to hold given the dataset size.

Contribution to open science in Central Asia. The complete pipeline - API ingest, SQLite storage, feature construction, model training, evaluation, and Streamlit deployment - is version-controlled in a single Python repository with a fully locked dependency specification (`uv.lock`) and `pyproject.toml`. Any researcher can rebuild the dataset, retrain the models, and redeploy the application using only a personal OpenAQ API key and a laptop with internet access. The pipeline’s configuration is centralised in `src/config.py`, where the city centre coordinates, training window, and API parameters are set; changing these four values is the only modification required to adapt the pipeline to another Central Asian city. This design choice - prioritising portability over technical sophistication - reflects a deliberate positioning of the work as a practical baseline for the region rather than a methodological benchmark for the global forecasting literature.

Future work should prioritise two directions. The first is LCS bias correction via a short co-location campaign alongside a Kazhydromet reference instrument, which would improve the absolute accuracy of the training targets and enable comparison with published RMSE figures from reference-grade studies. The second is integration of the Open-Meteo forecast BLH product - forecast rather than analysed boundary-layer height at the prediction time - which would extend the model’s look-ahead on inversion evolution and partially compensate for the 2–3 hour lag at on-set transitions that is the principal operational weakness of the current architecture. A third direction of practical value is a weighted ensemble combining XGBoost’s 6-hour precision with Ridge regression’s 12 and 24-hour stability, which is straightforward to implement and expected to match or exceed both individual models at all horizons without additional training data. These three improvements would move the system from its current research-prototype status toward an operationally deployable public health tool for one of Central Asia’s most polluted cities.

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A. Source Code Excerpts